

Contribution Validation Report — ONE simulation run only

Case: all contributions matched

Type: Matched validation — all deterministic effects matched (expect mismatch < 0.05 m)

Duration: 600 s

Towers: 5

Receivers: 1

Max PR meas/epoch: 5

Max EKF rows/epoch: 5

Final pos error: 5.7890 m

Pos RMS (last 20%): 5.0624 m

Mean NIS: 5.02 (E[NIS] ~ 5)

Truth effects: Sagnac, Shapiro, Trop, Iono, TowerSurvey, PCO, PCV

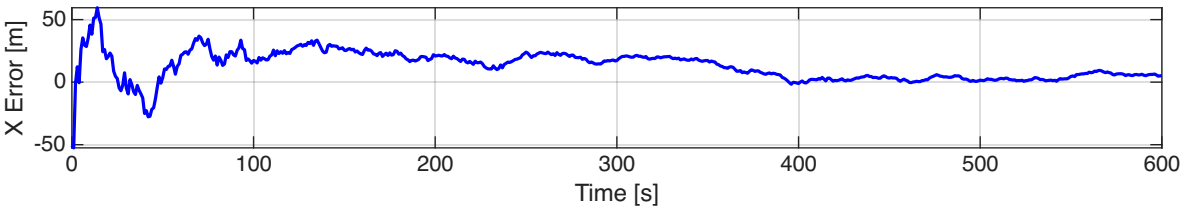
Model effects: Sagnac, Shapiro, Trop, Iono, TowerSurvey, PCO, PCV

Mismatched: none

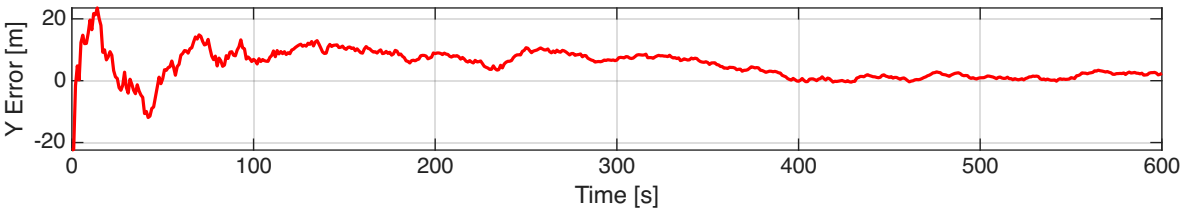
Output PDF: /Users/ludwigmatuschka/Library/CloudStorage/OneDrive-Persönlich/Dokumente/ICH/Karriere/Cranfield/IRP/Codes/IRP_simulation/oo_v1/output/effect_contribution_validation_report.pdf

Position Error Components (ECEF)

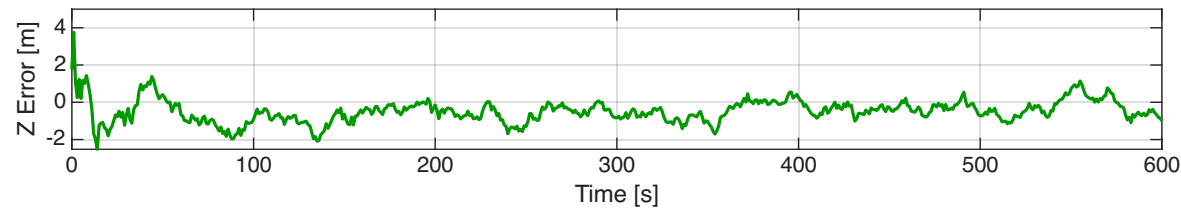
ECEF Position Error — X



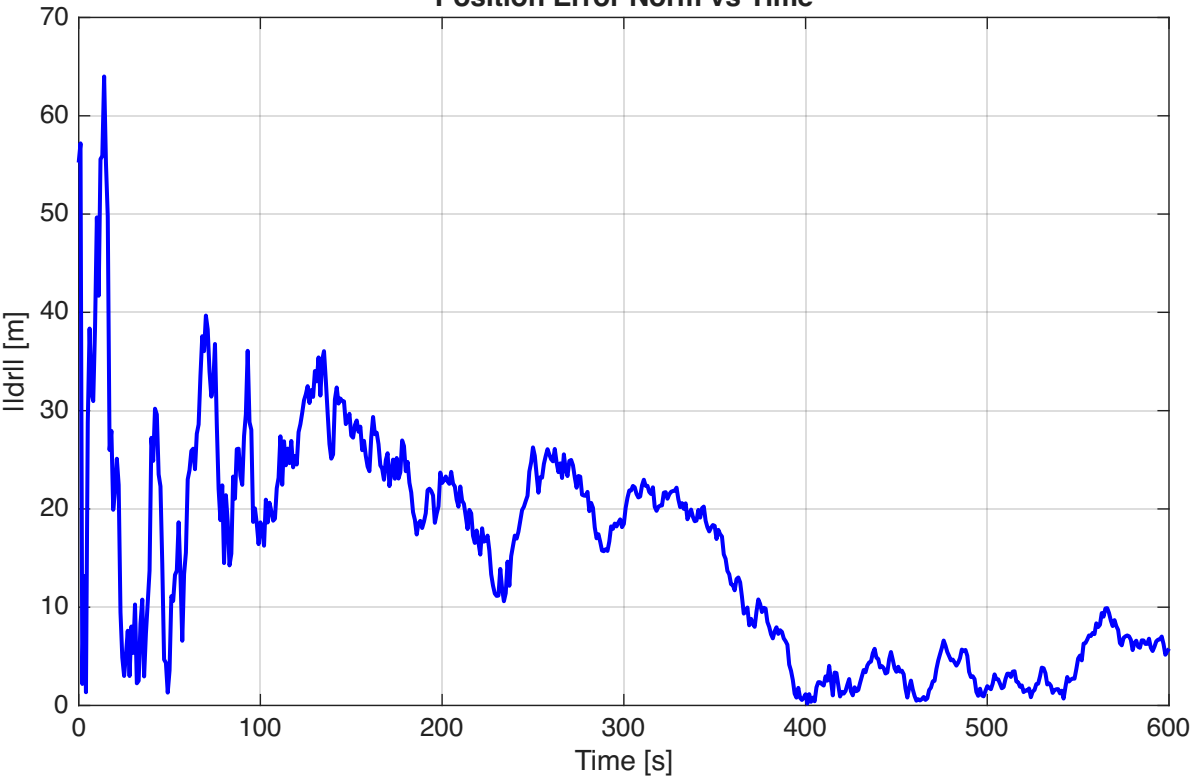
ECEF Position Error — Y



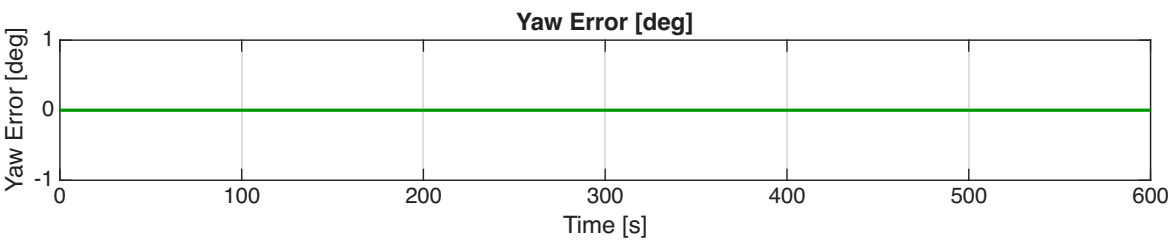
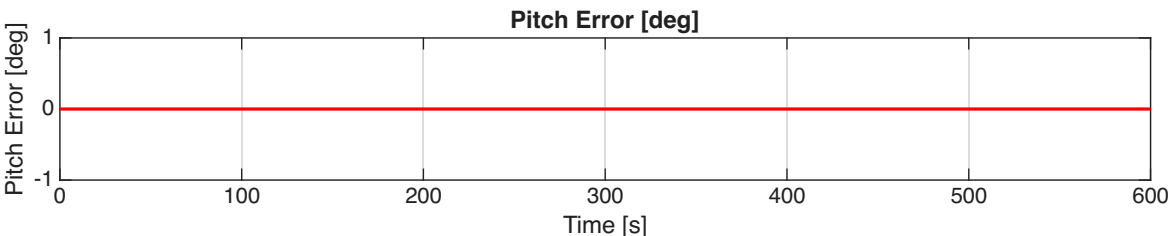
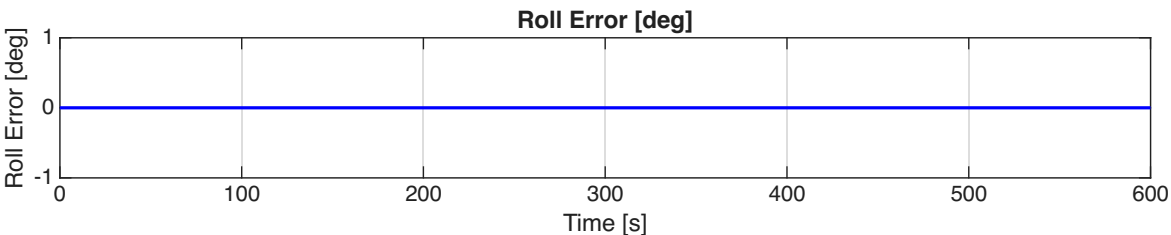
ECEF Position Error — Z



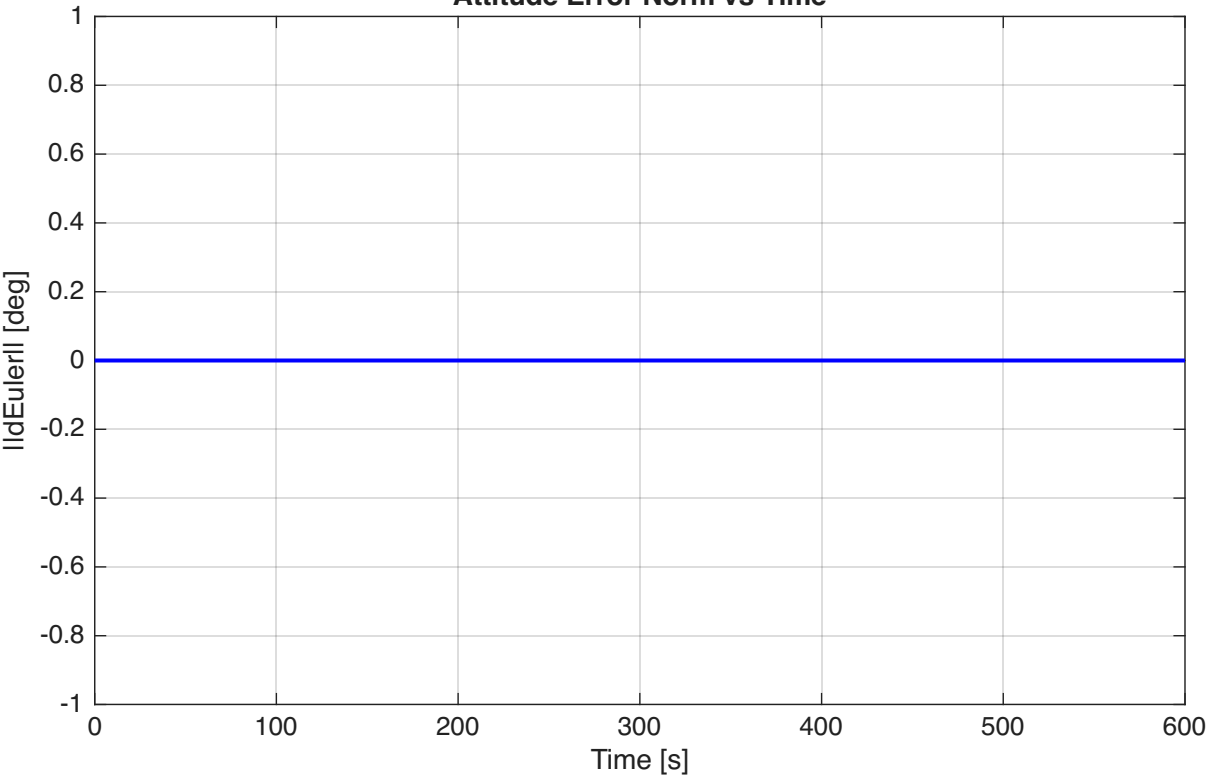
Position Error Norm vs Time



Attitude Error Components

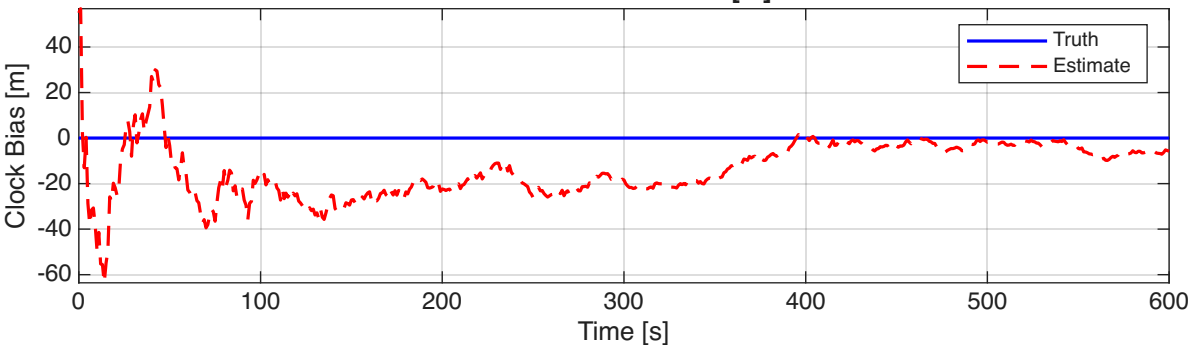


Attitude Error Norm vs Time

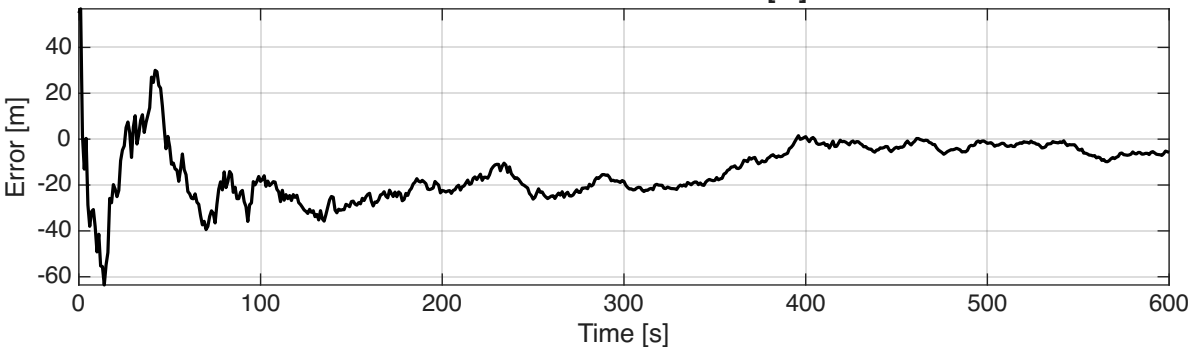


Receiver Clock Bias

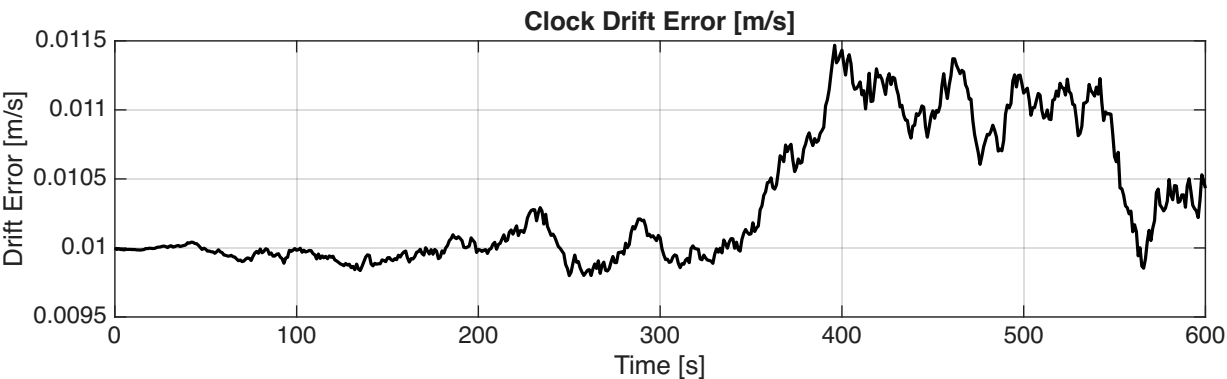
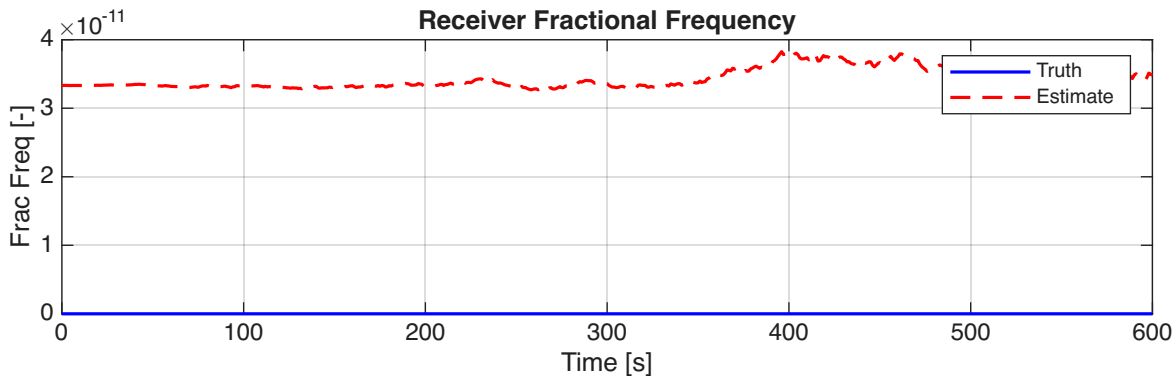
Receiver Clock Bias [m]



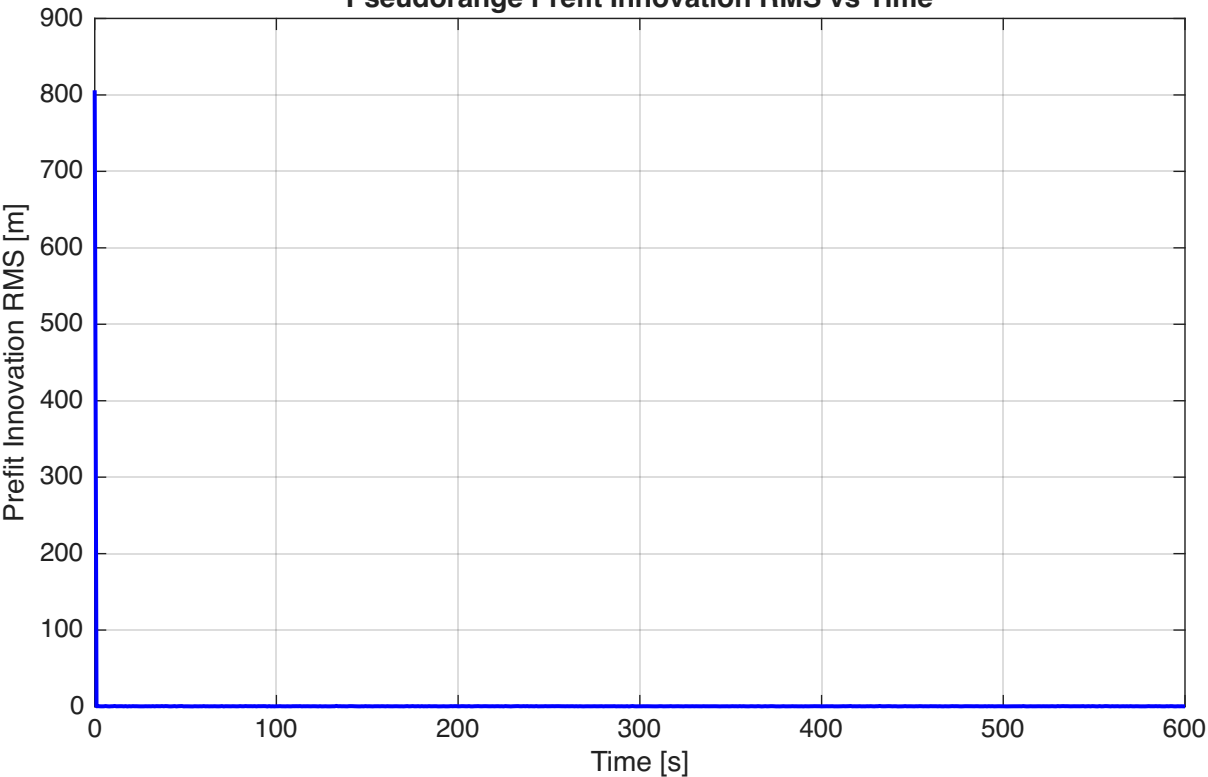
Receiver Clock Bias Error [m]



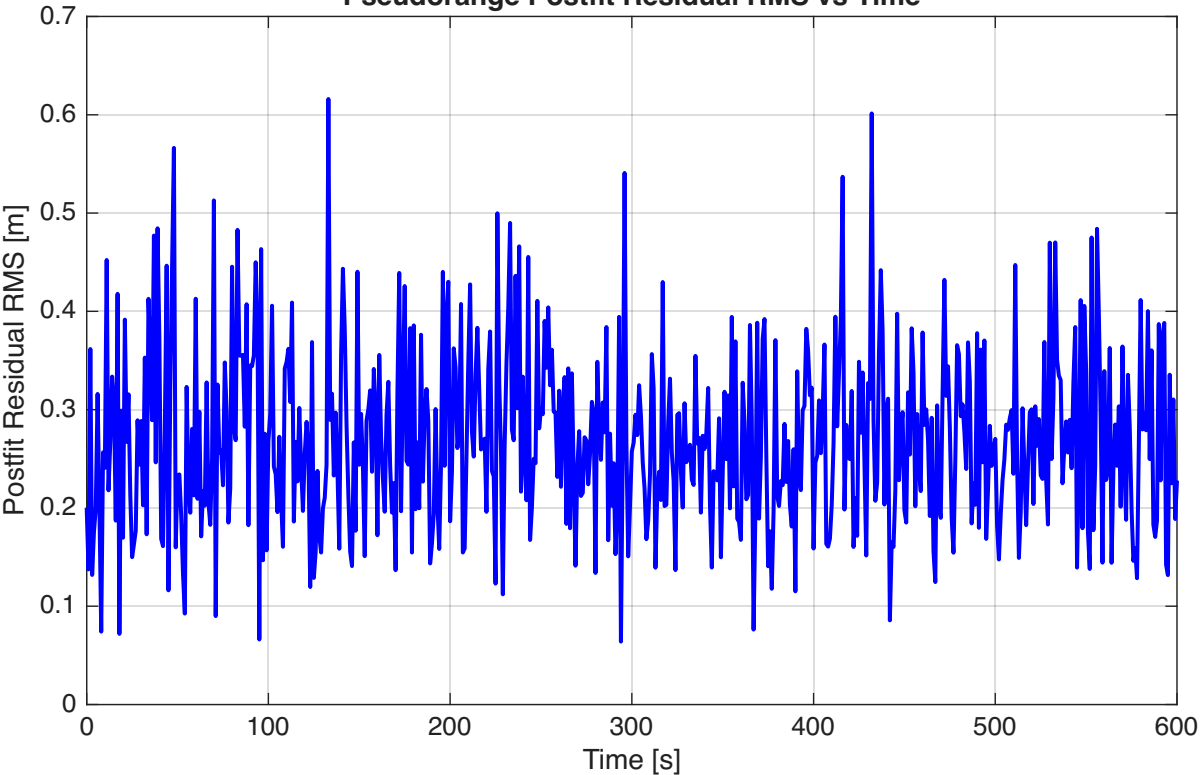
Receiver Clock Drift / Fractional Frequency



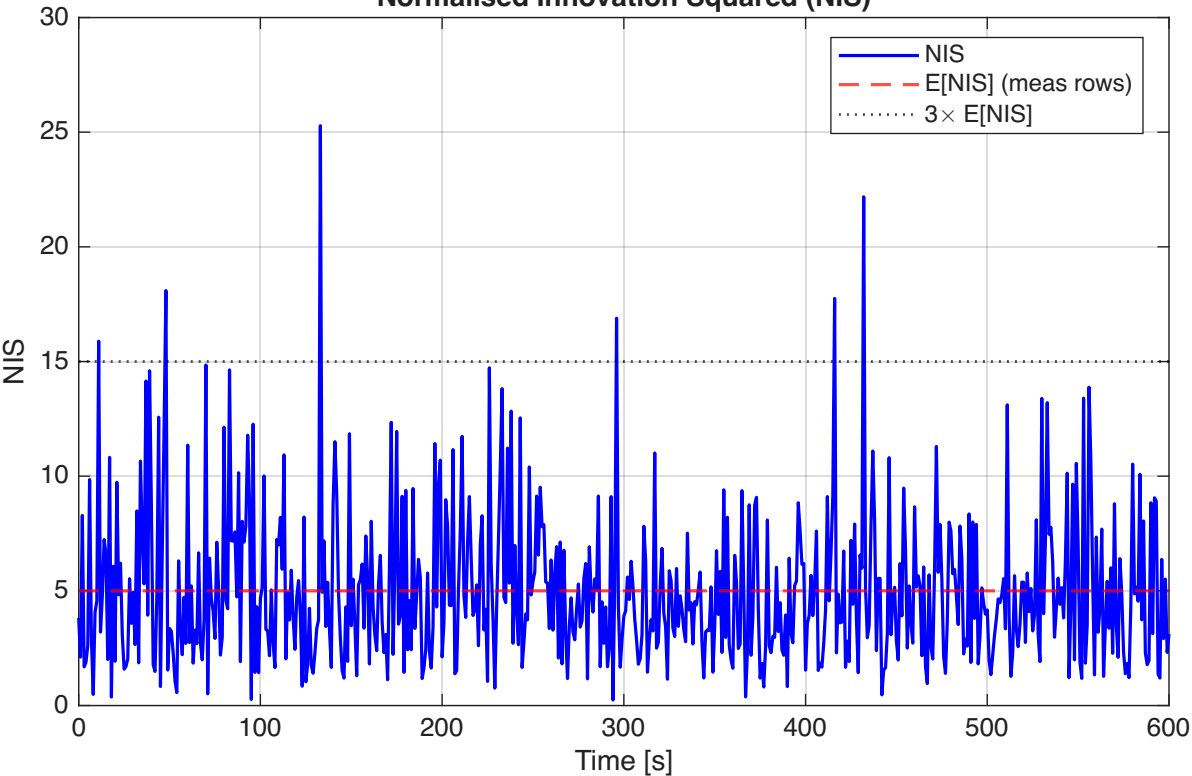
Pseudorange Prefit Innovation RMS vs Time



Pseudorange Postfit Residual RMS vs Time

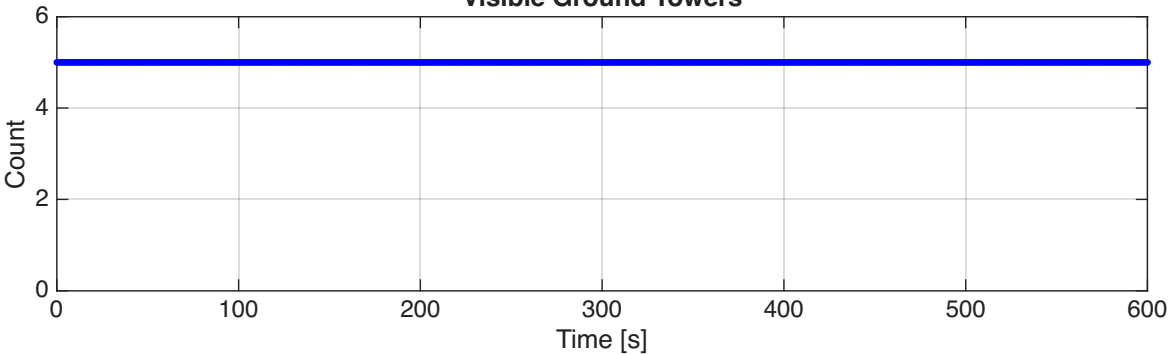


Normalised Innovation Squared (NIS)

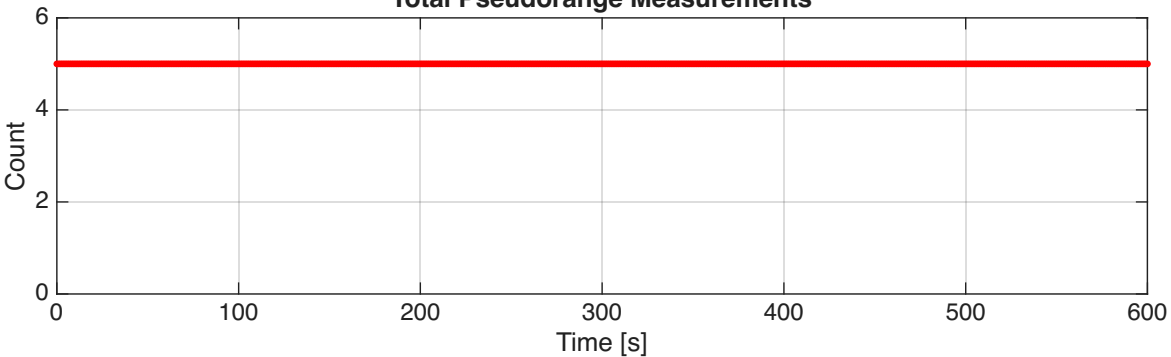


Observations per Epoch

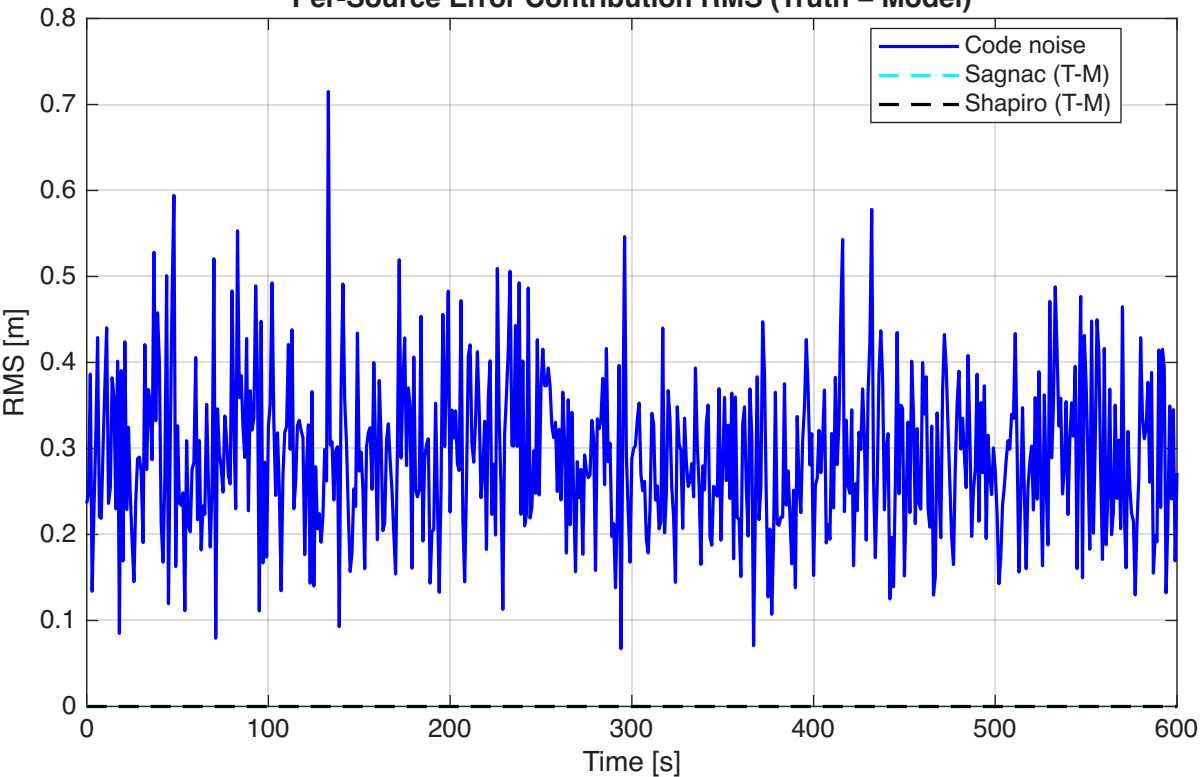
Visible Ground Towers



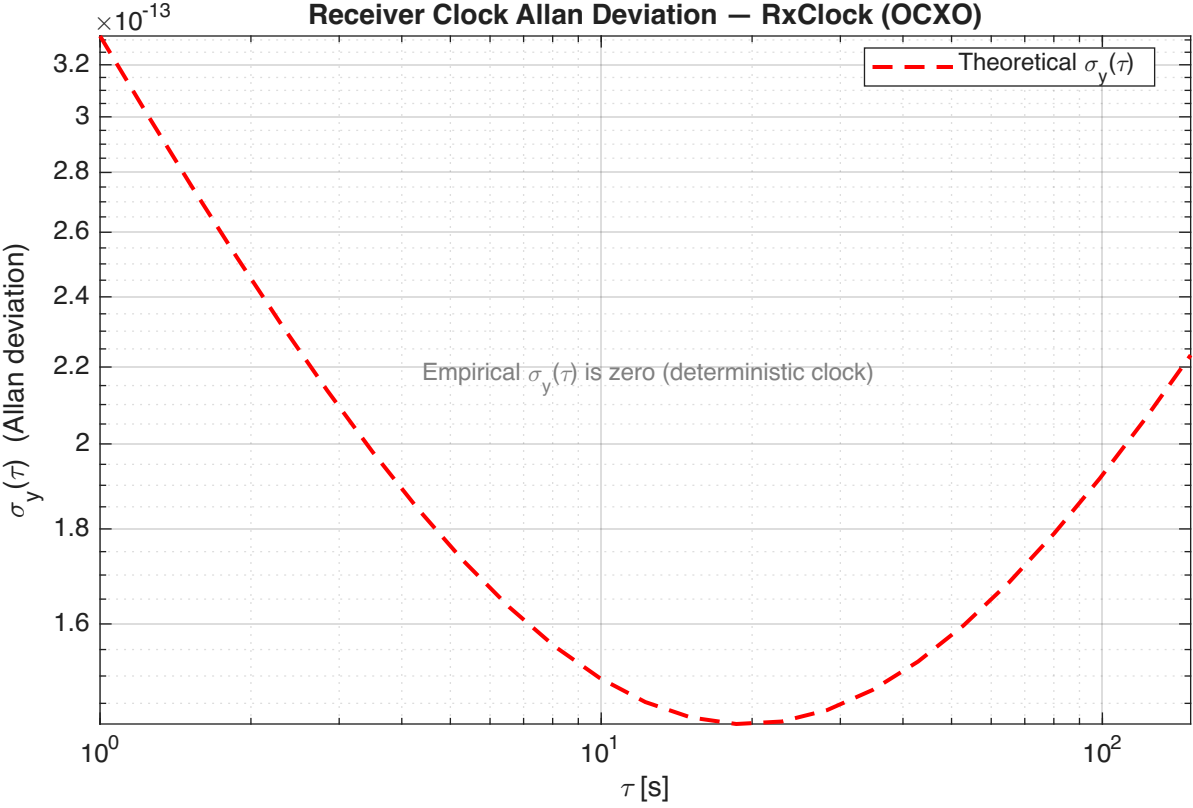
Total Pseudorange Measurements



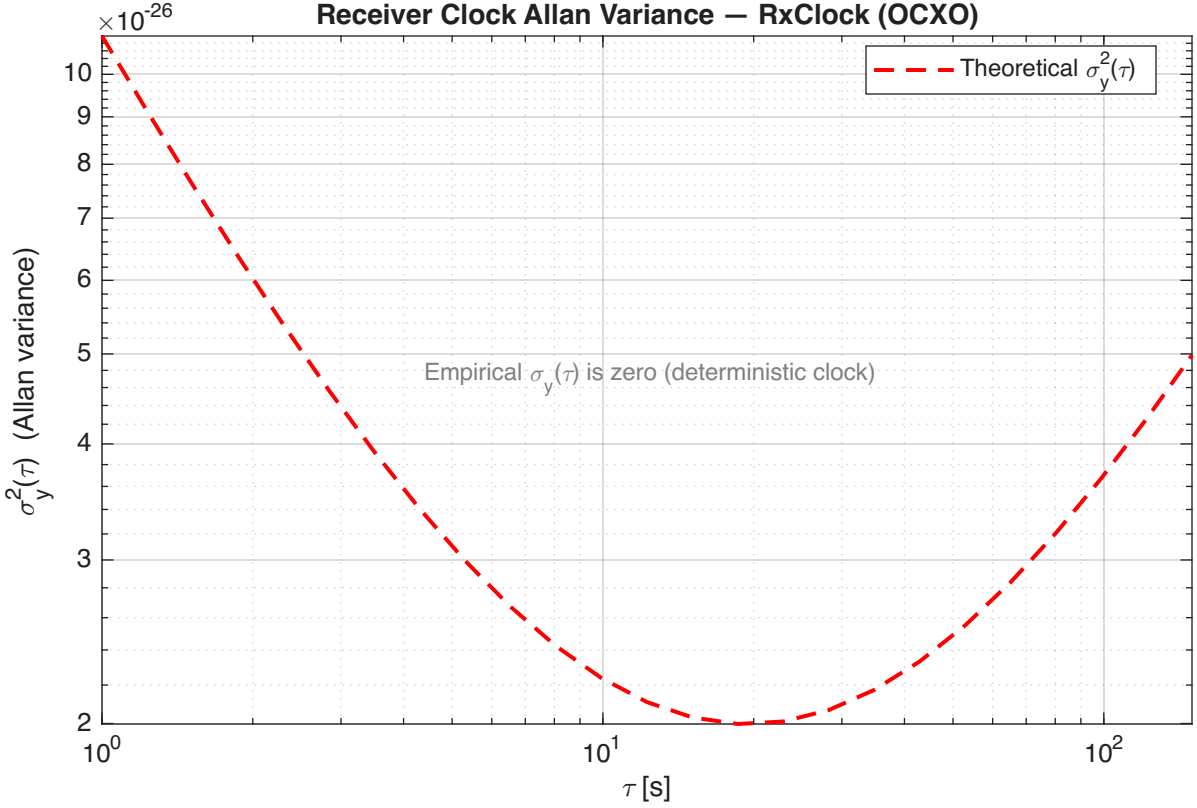
Per-Source Error Contribution RMS (Truth - Model)



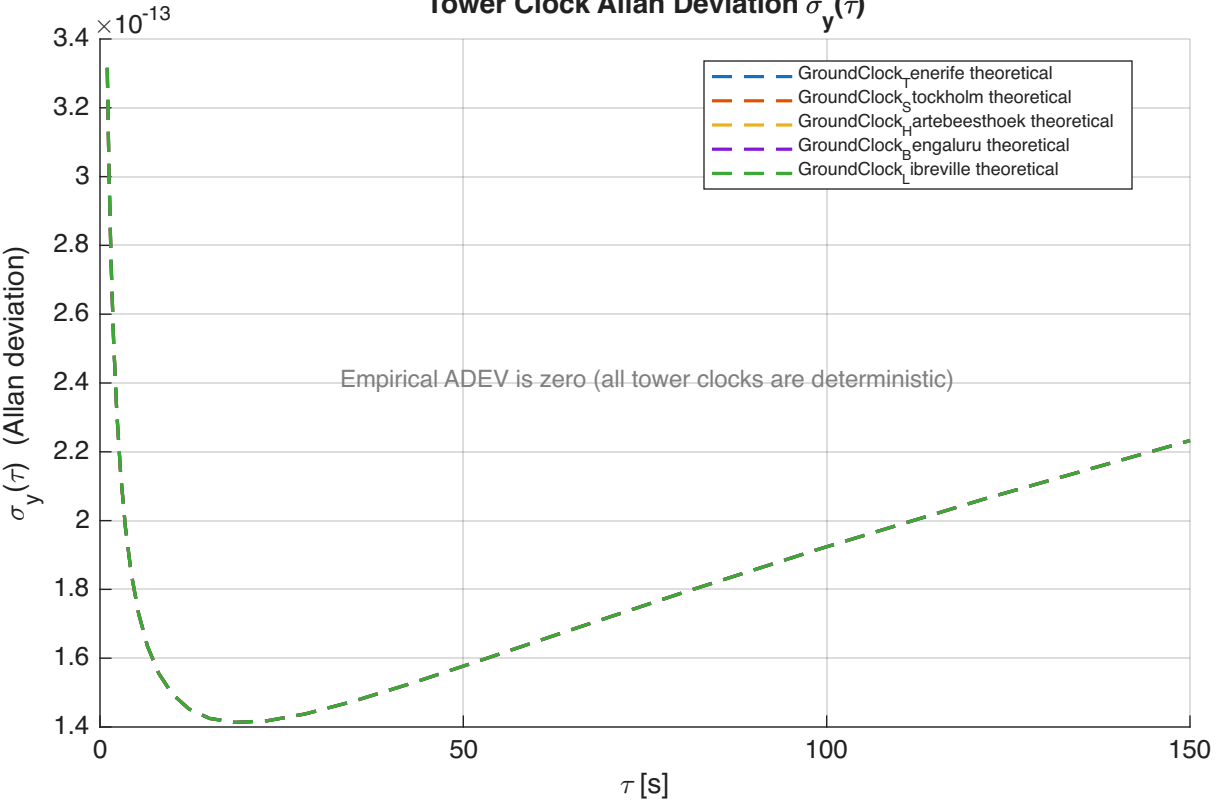
Receiver Clock Allan Deviation — RxClock (OCXO)



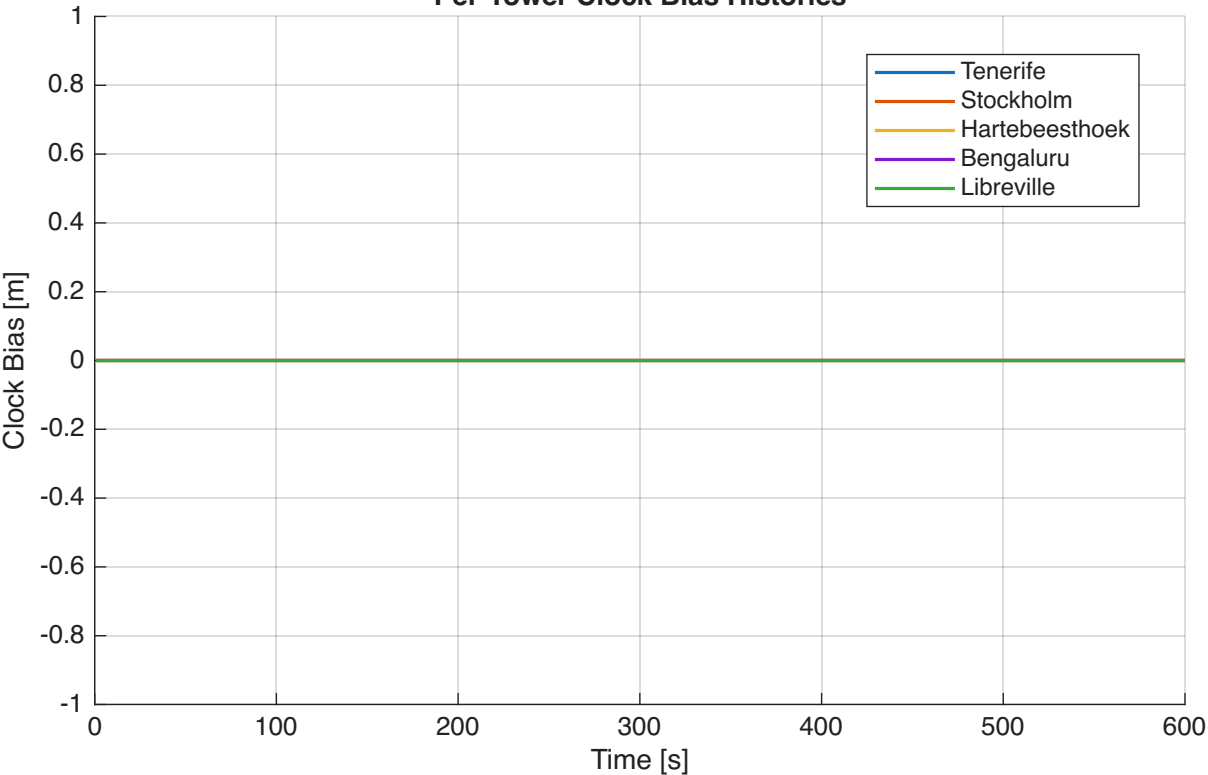
Receiver Clock Allan Variance — RxClock (OCXO)



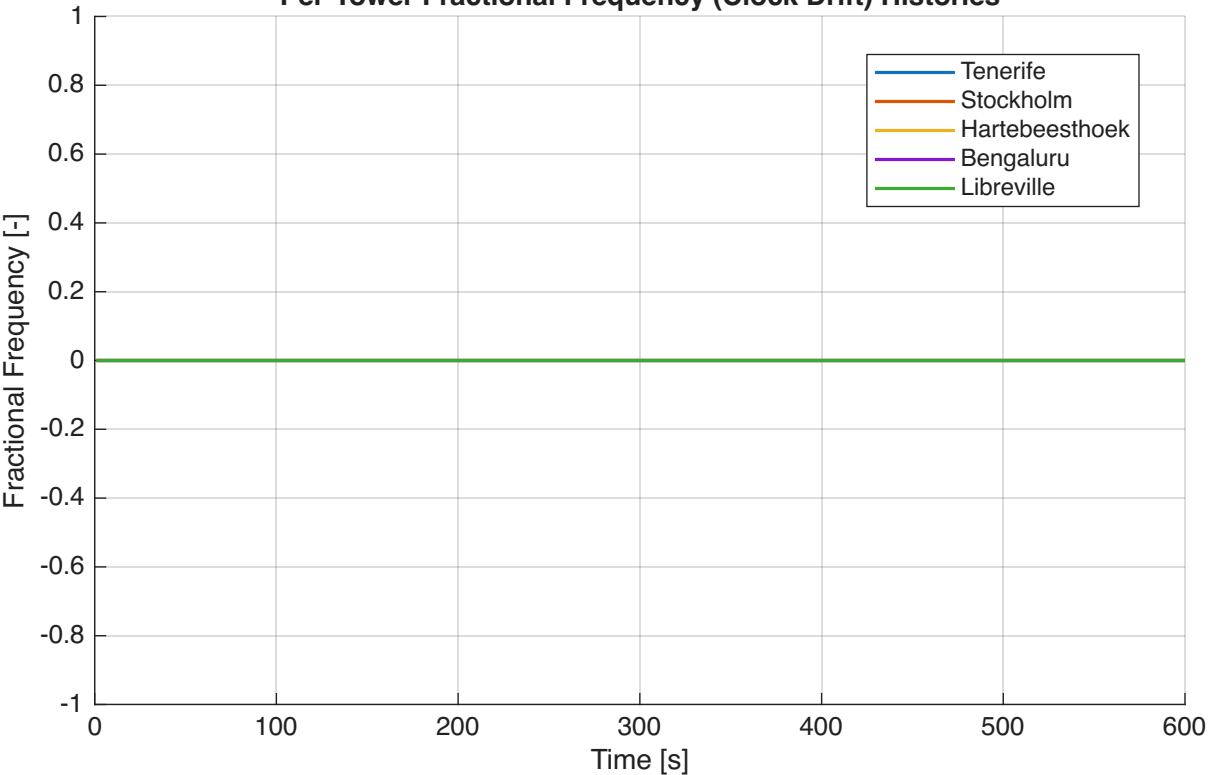
Tower Clock Allan Deviation $\sigma_y(\tau)$



Per-Tower Clock Bias Histories

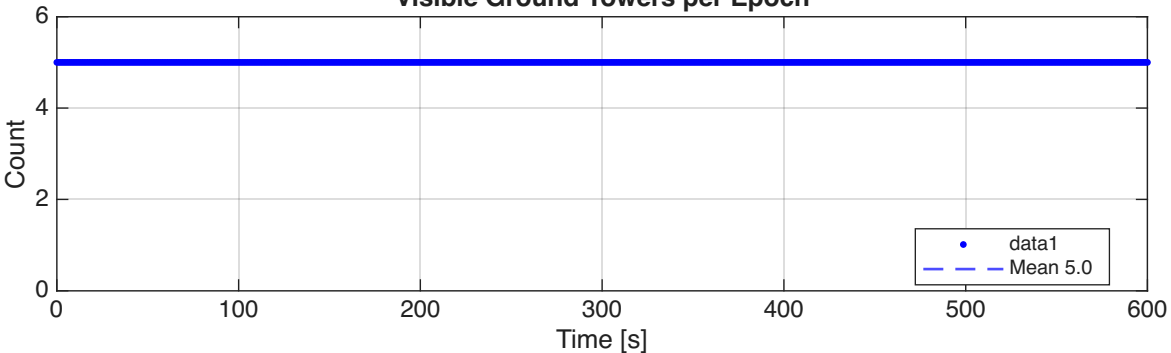


Per-Tower Fractional Frequency (Clock Drift) Histories

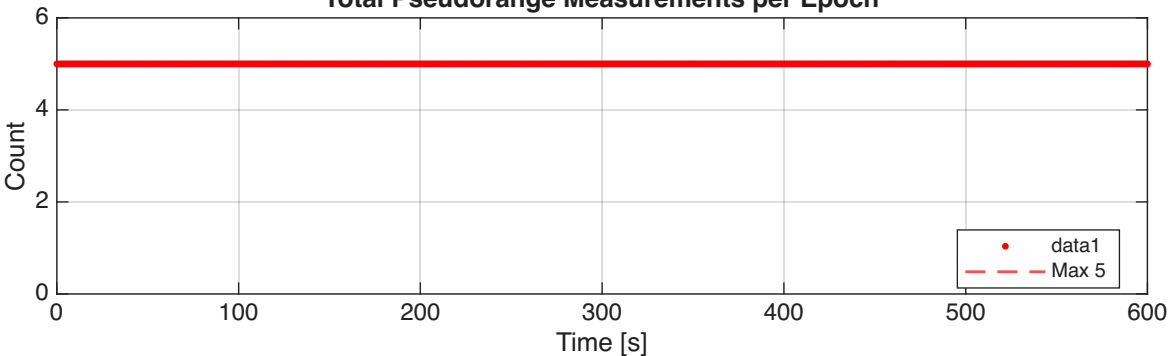


Observation Count (max meas = 5)

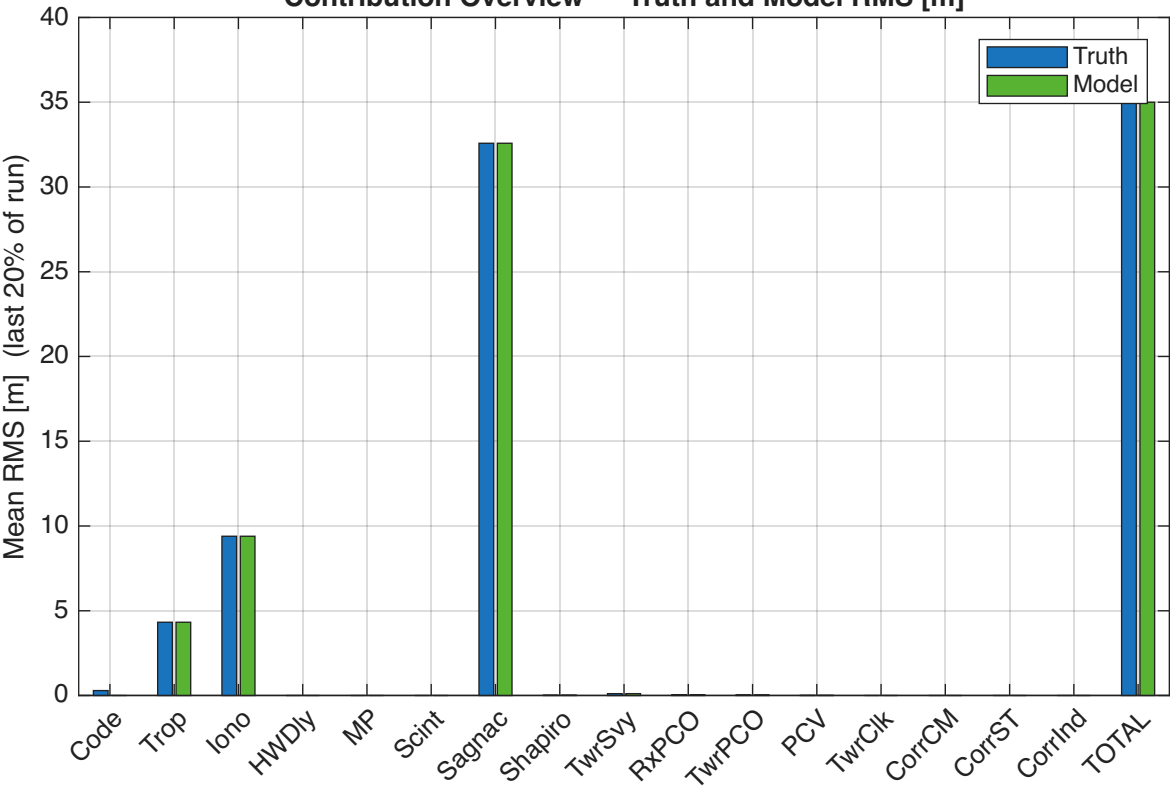
Visible Ground Towers per Epoch



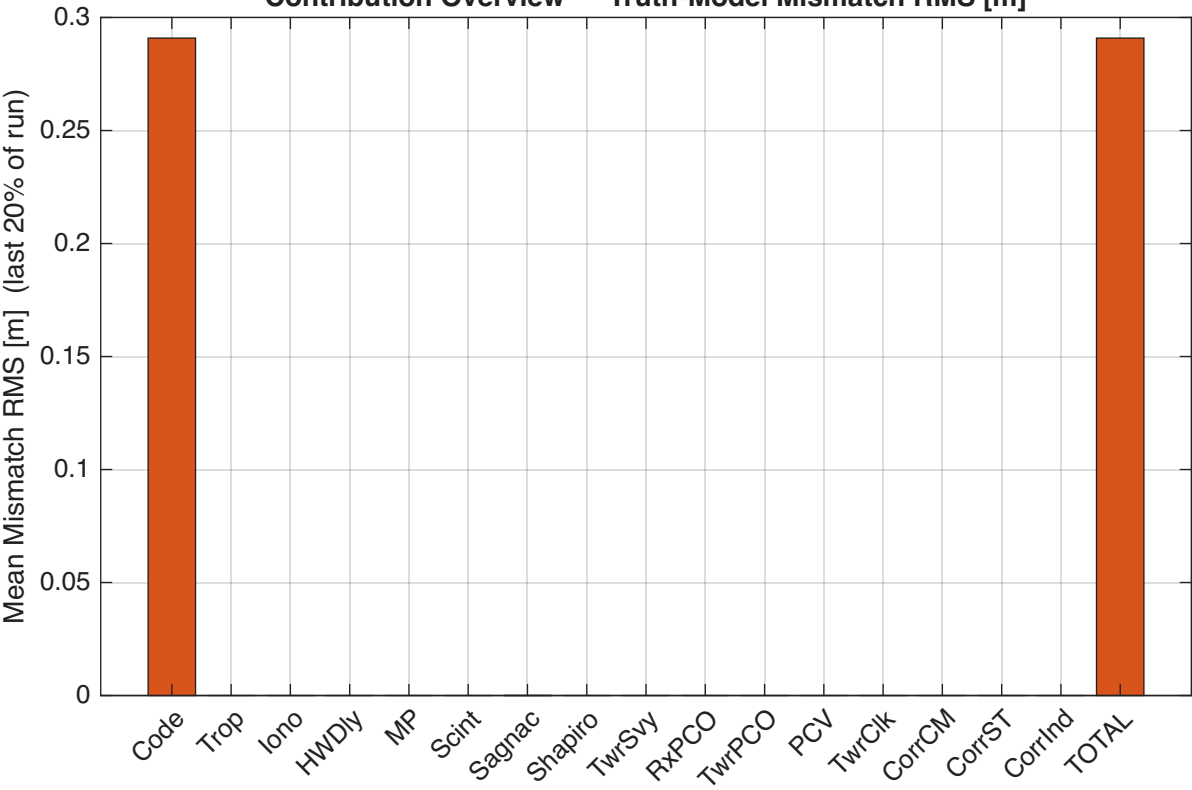
Total Pseudorange Measurements per Epoch



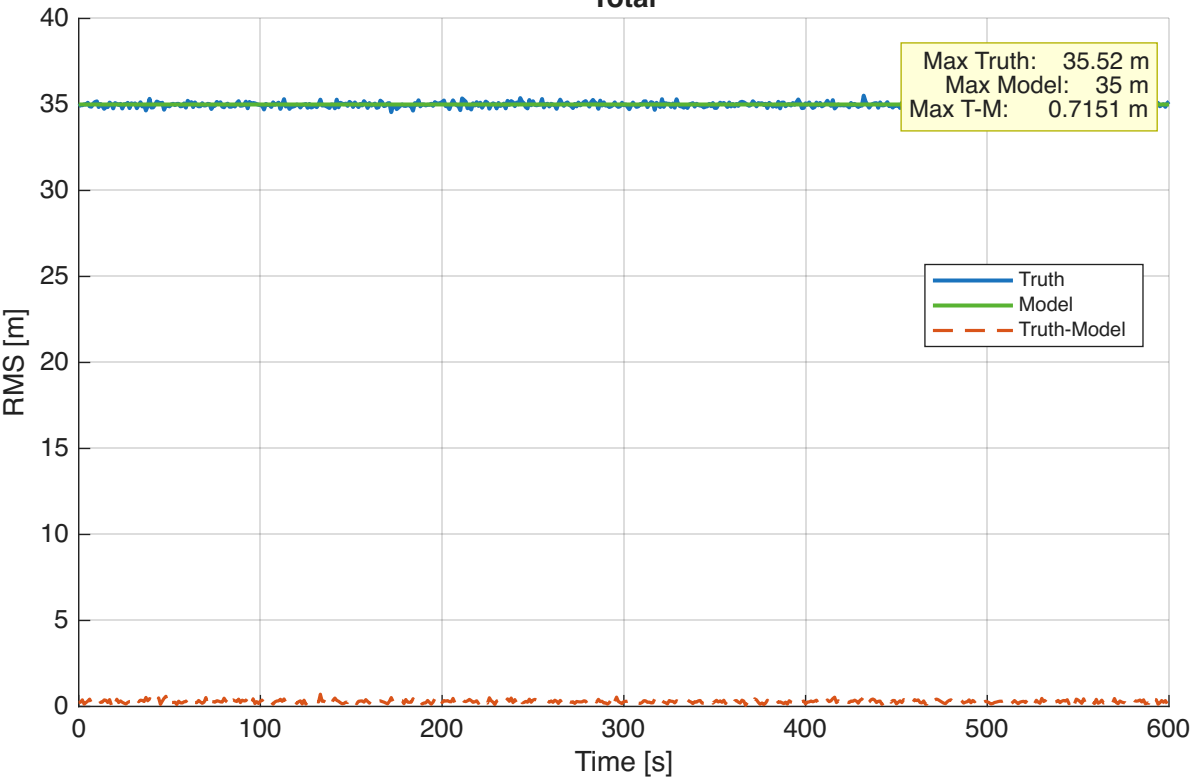
Contribution Overview — Truth and Model RMS [m]



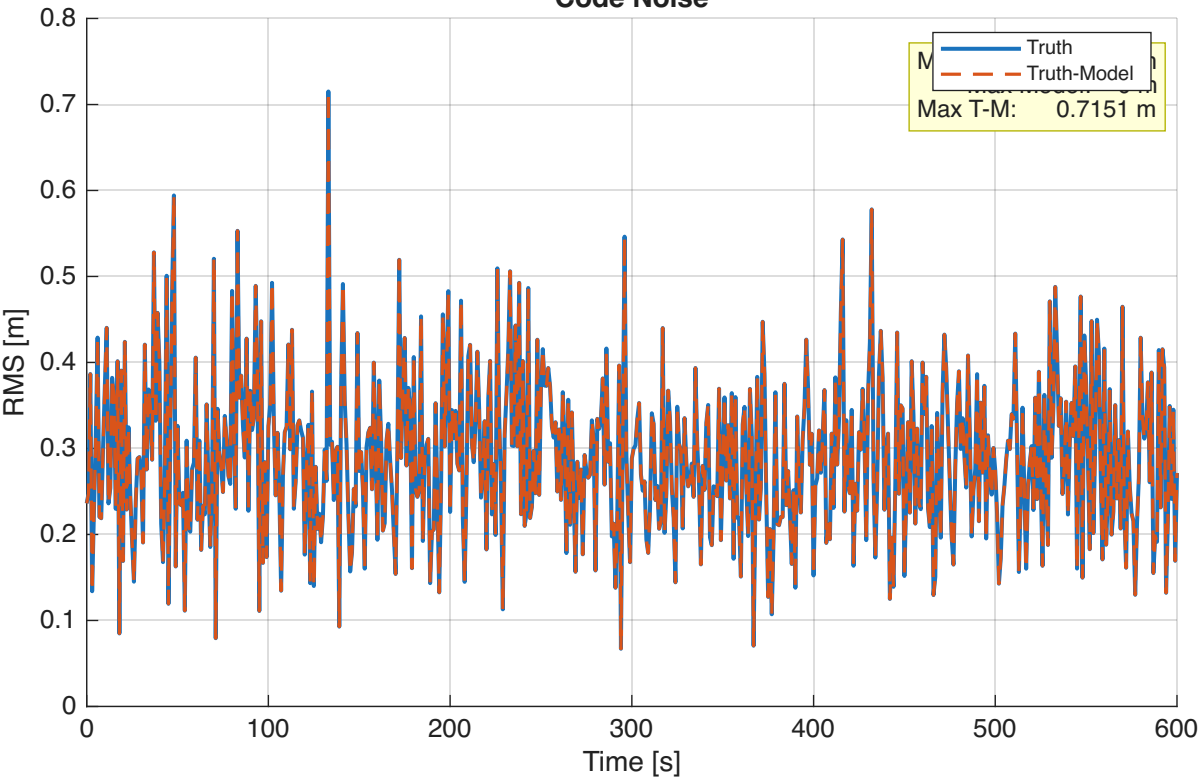
Contribution Overview — Truth-Model Mismatch RMS [m]



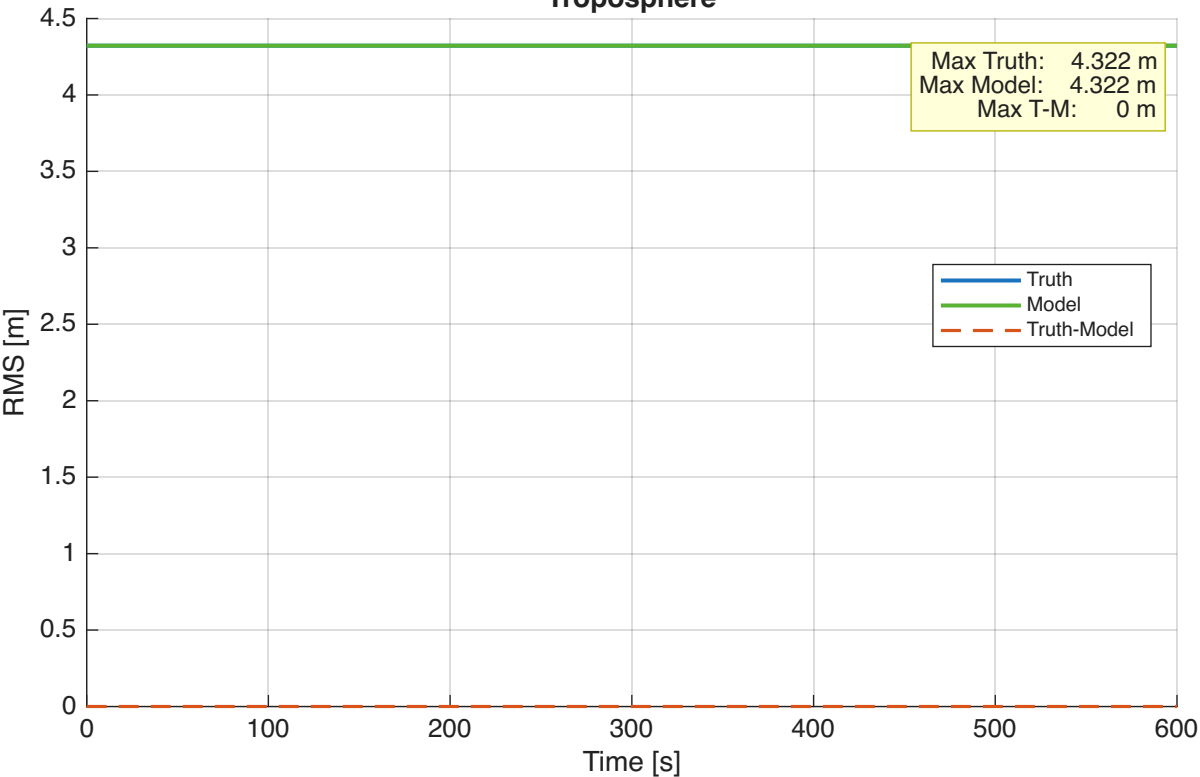
Total



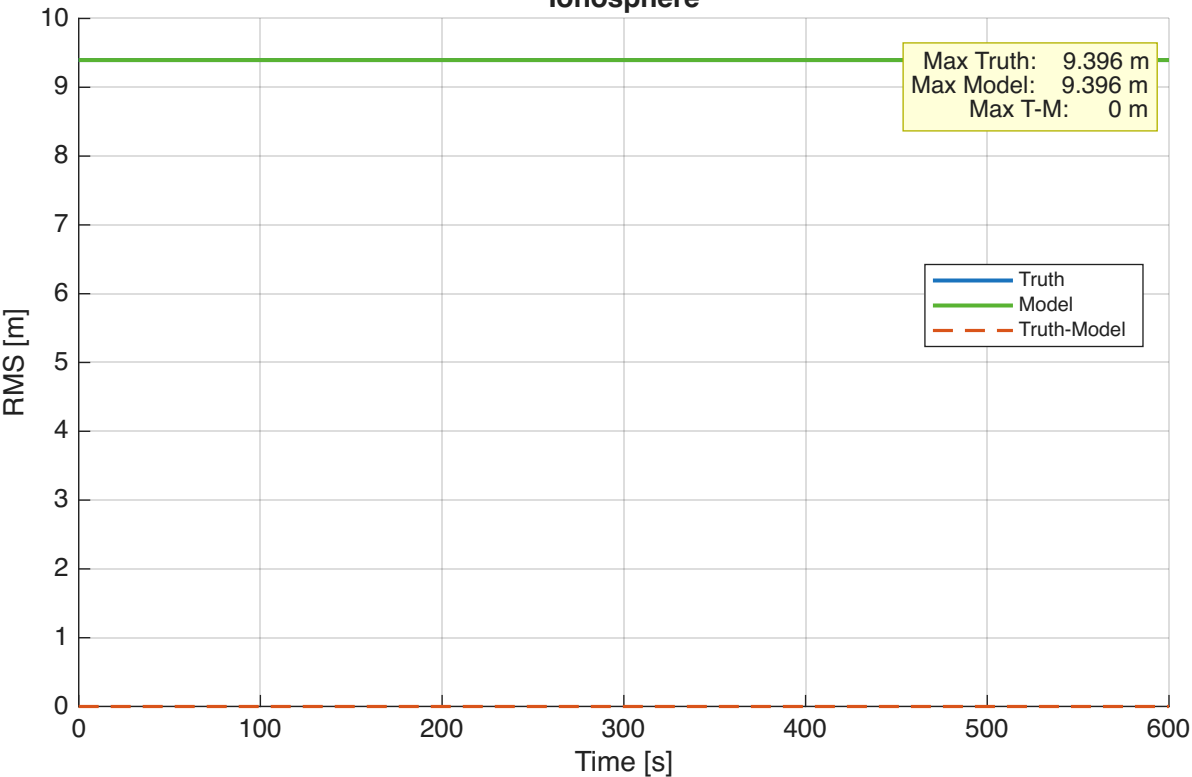
Code Noise



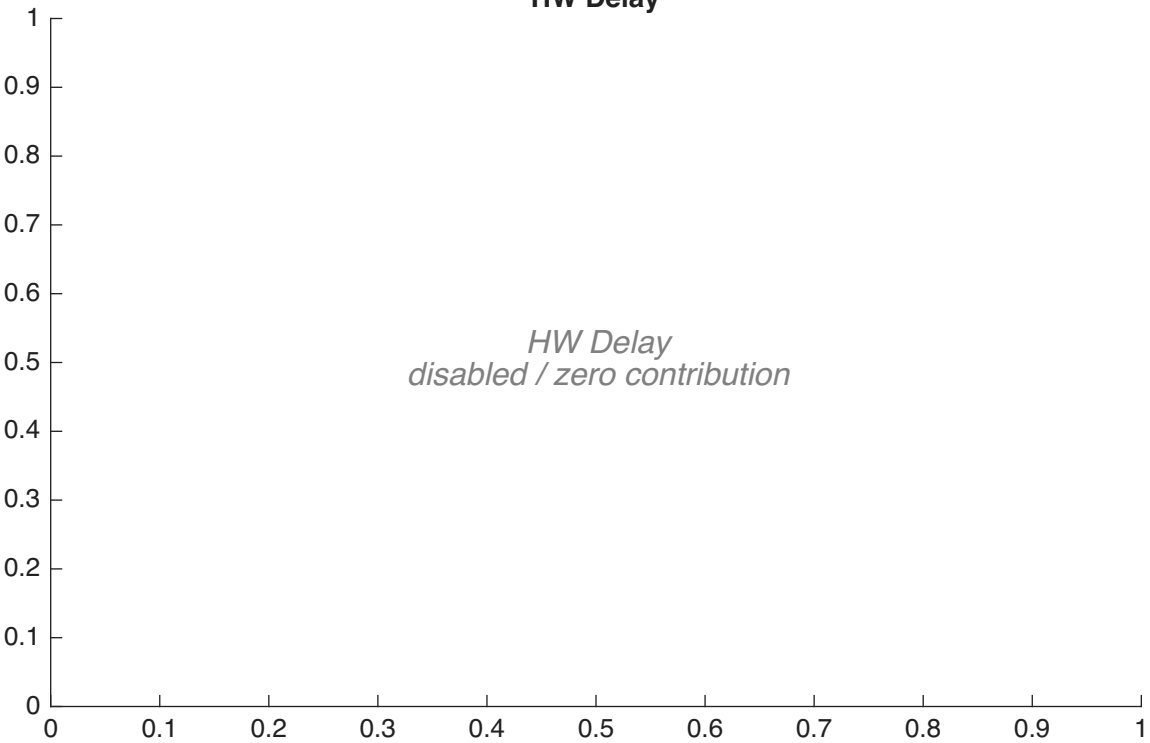
Troposphere



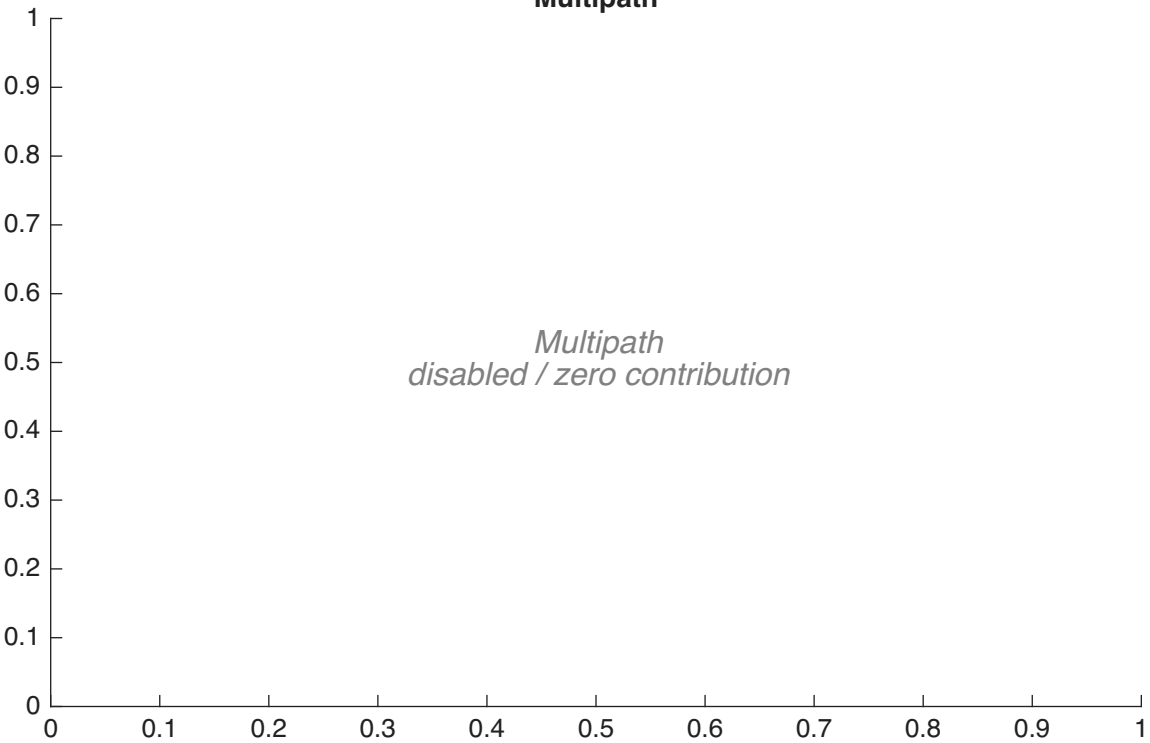
Ionosphere



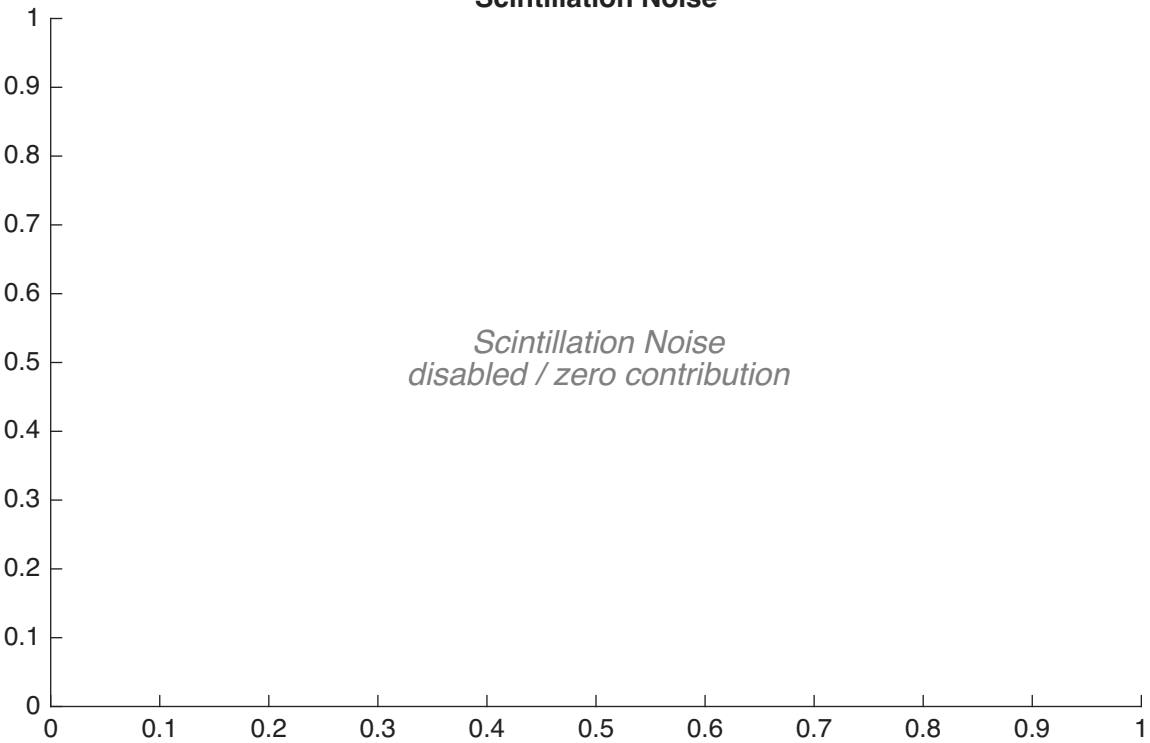
HW Delay



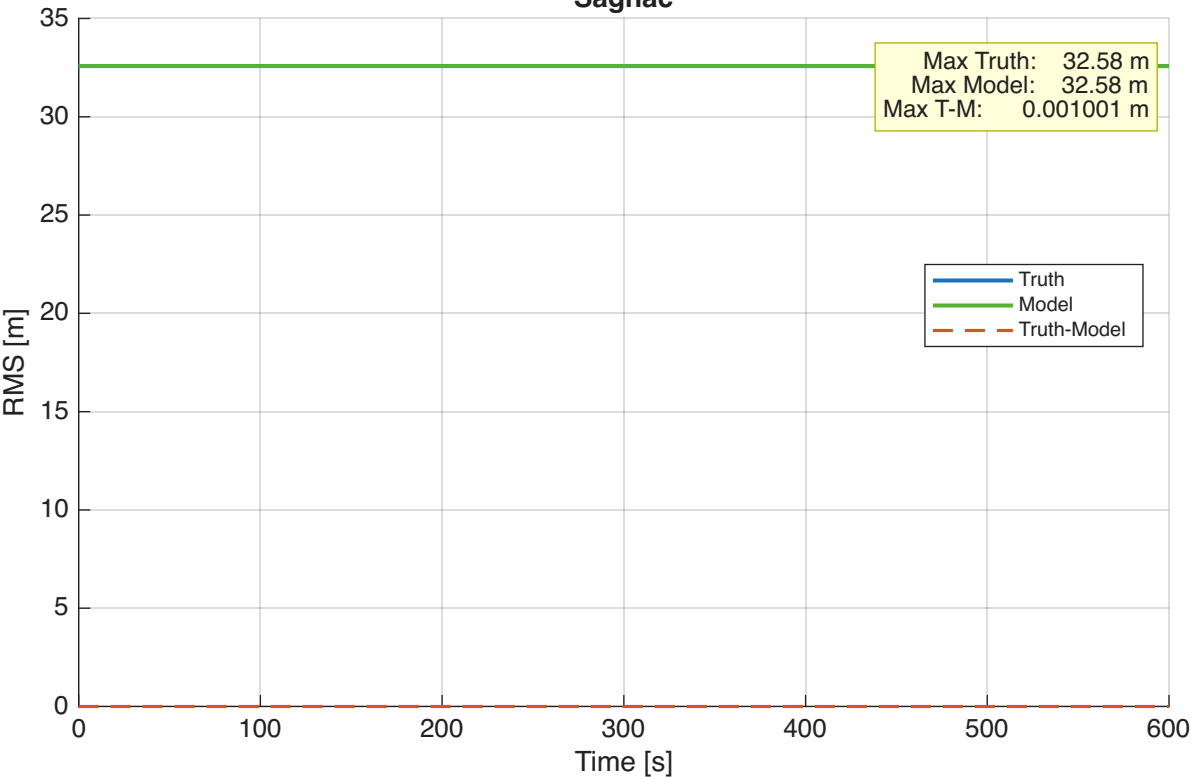
Multipath



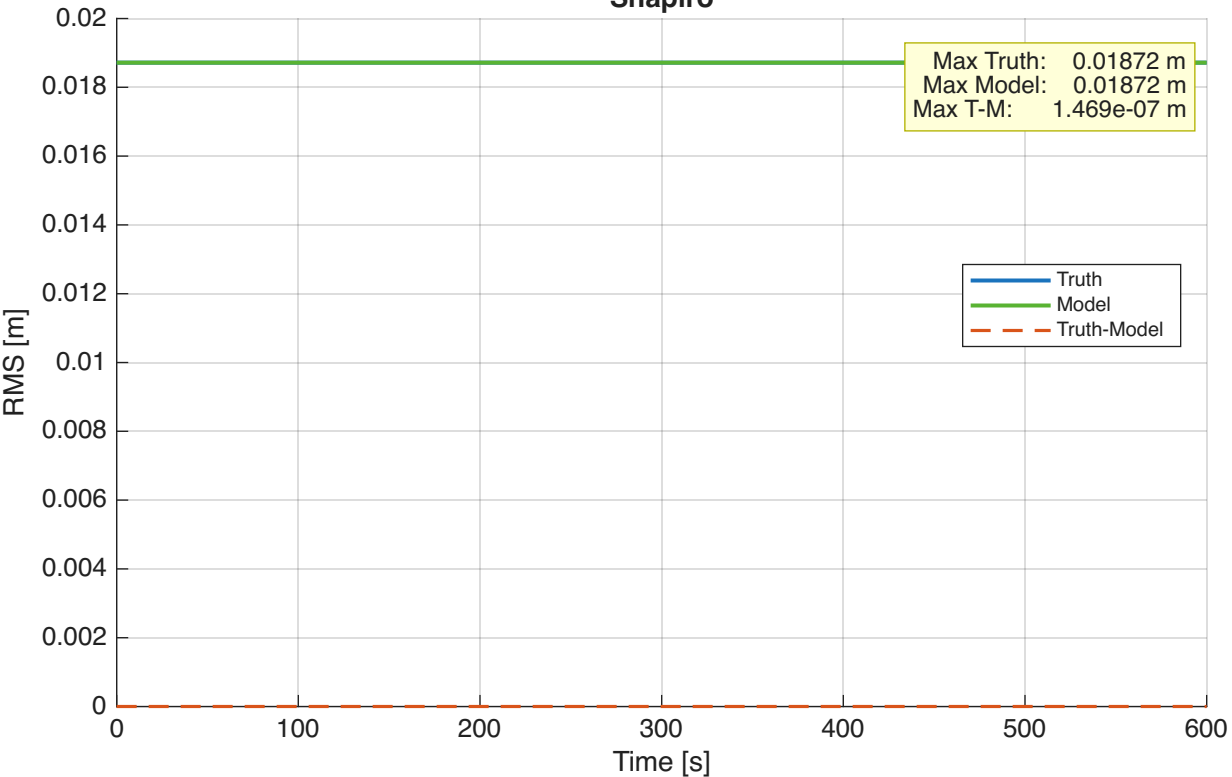
Scintillation Noise



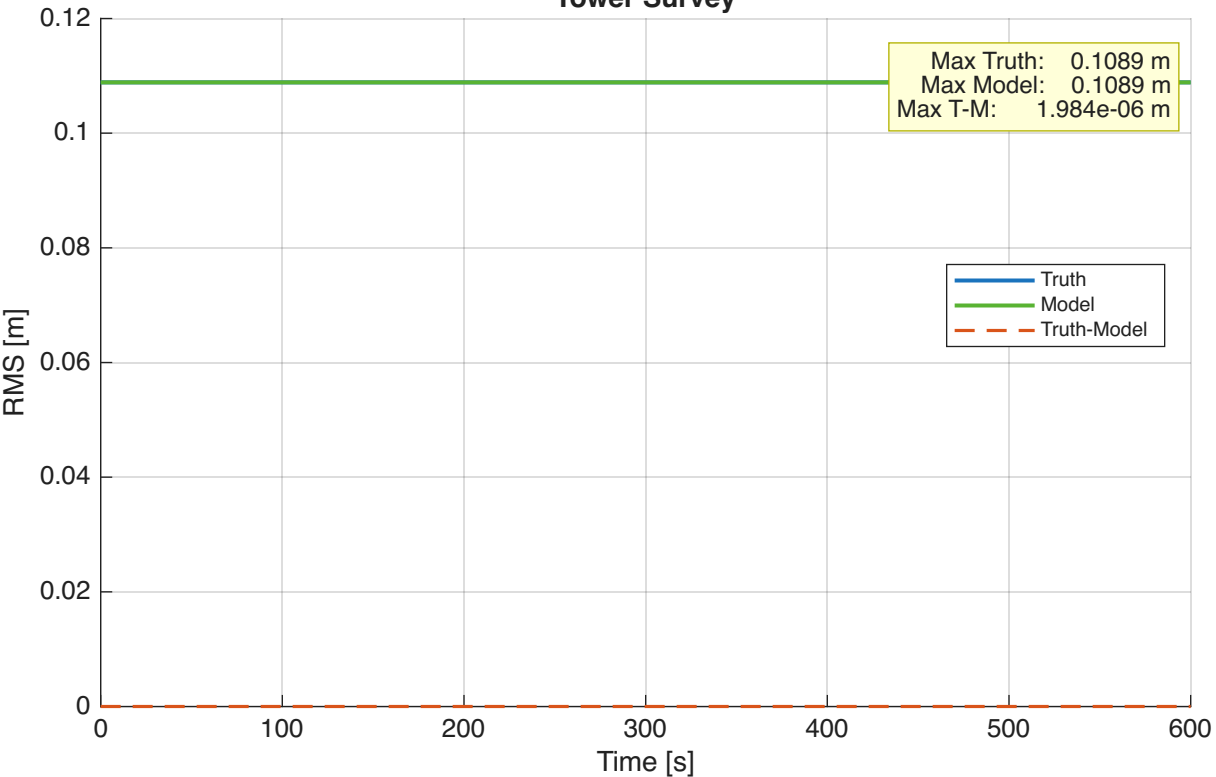
Sagnac



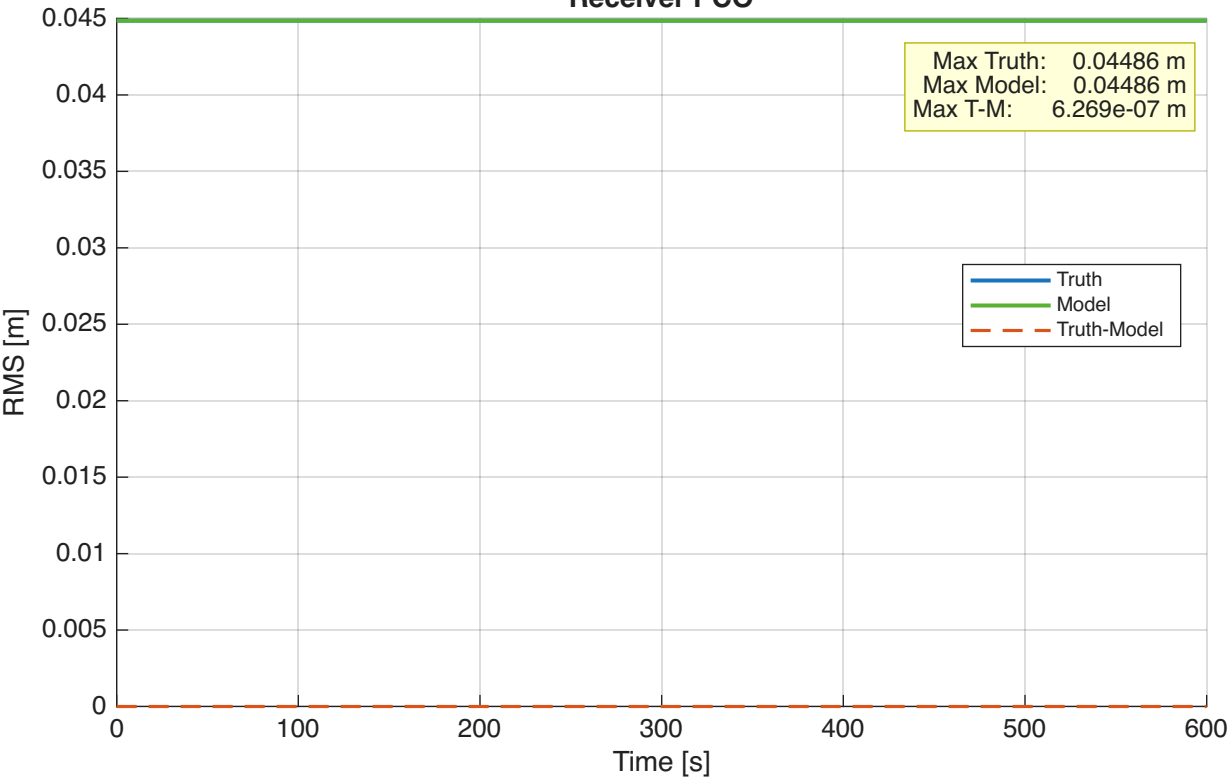
Shapiro



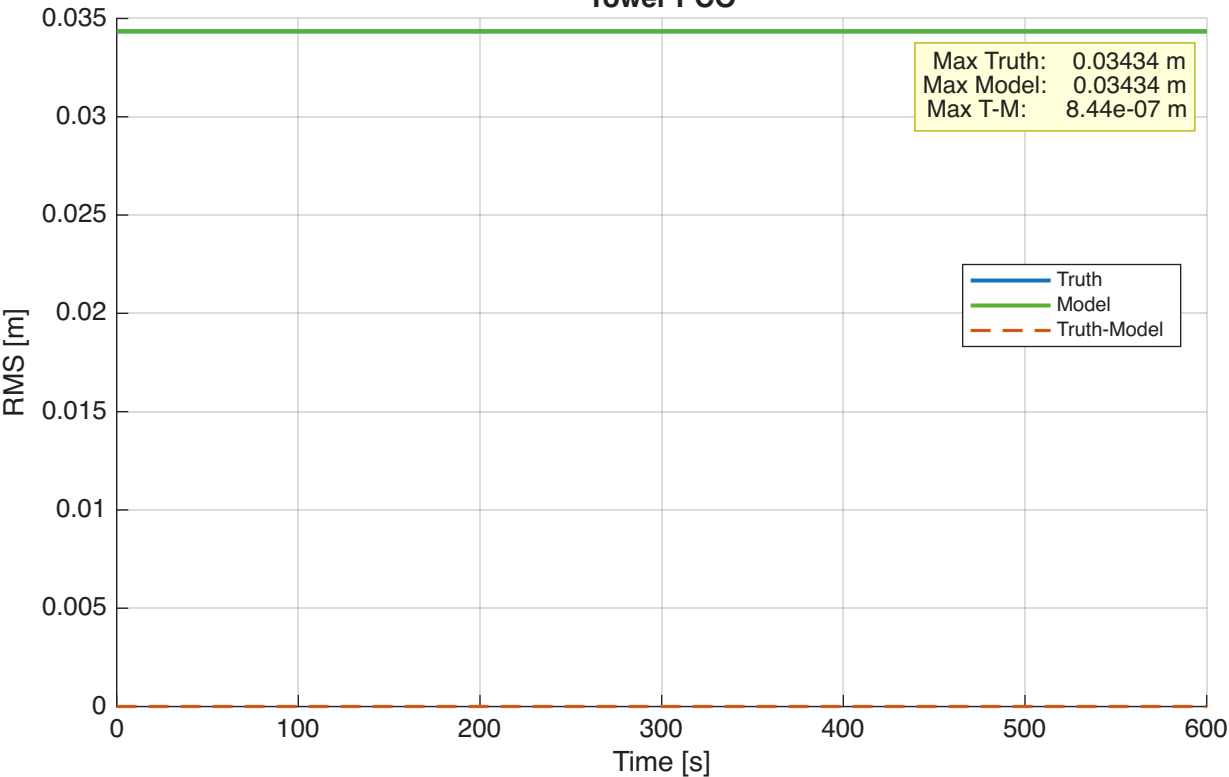
Tower Survey

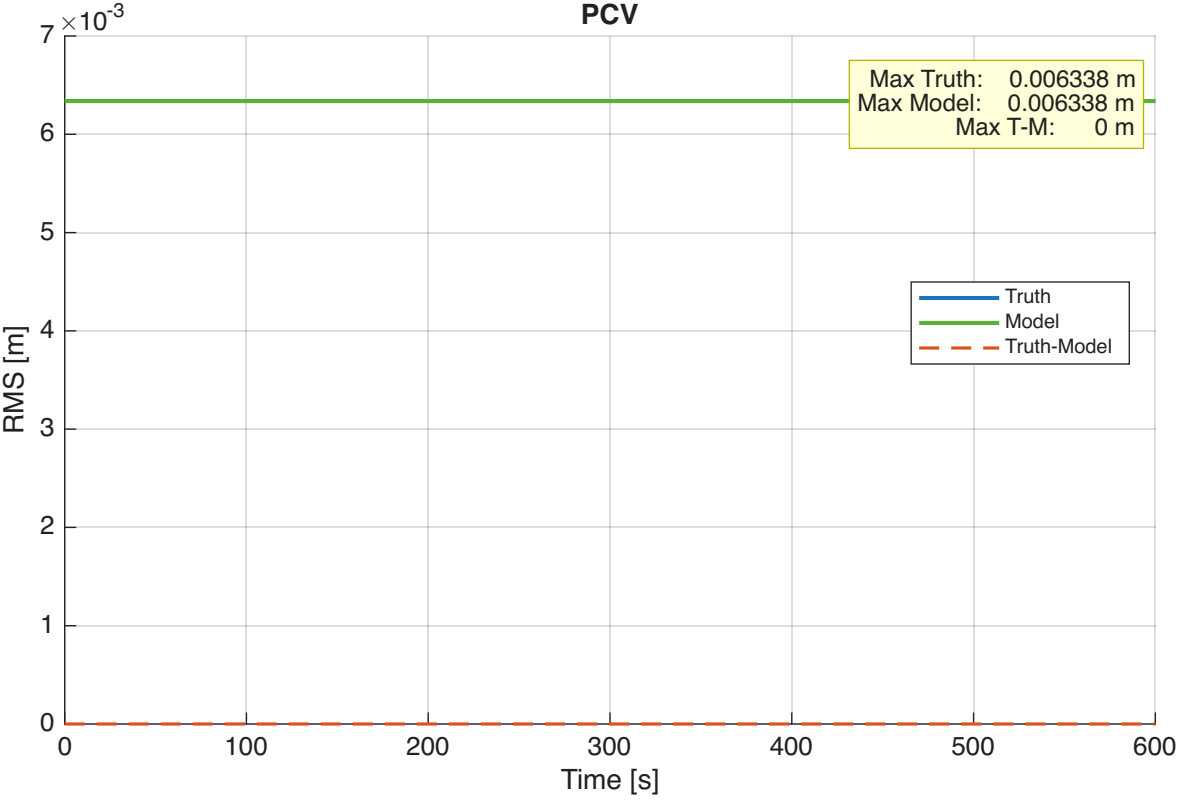


Receiver PCO

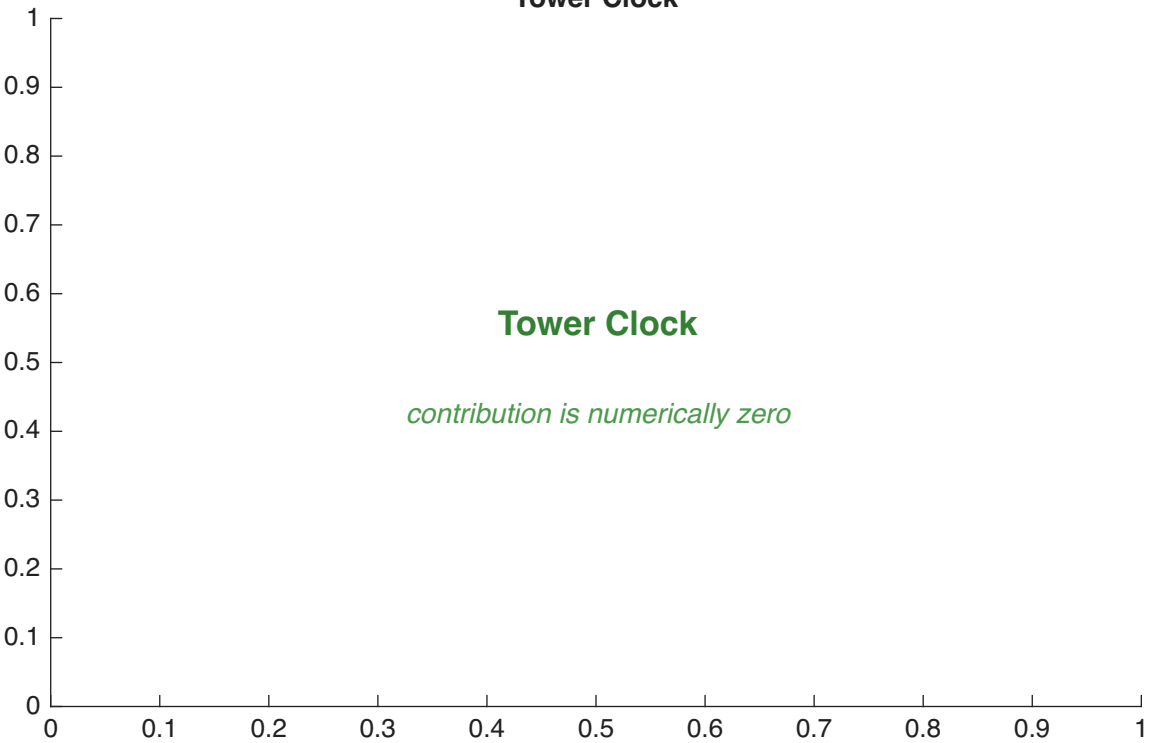


Tower PCO





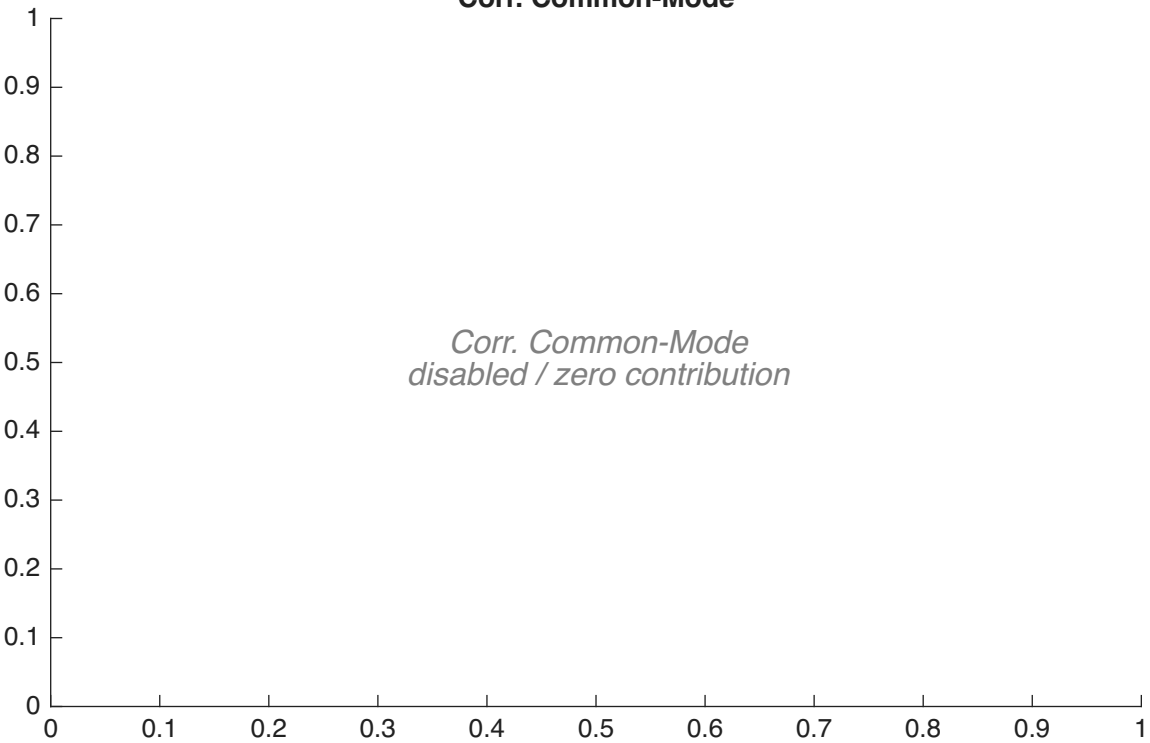
Tower Clock



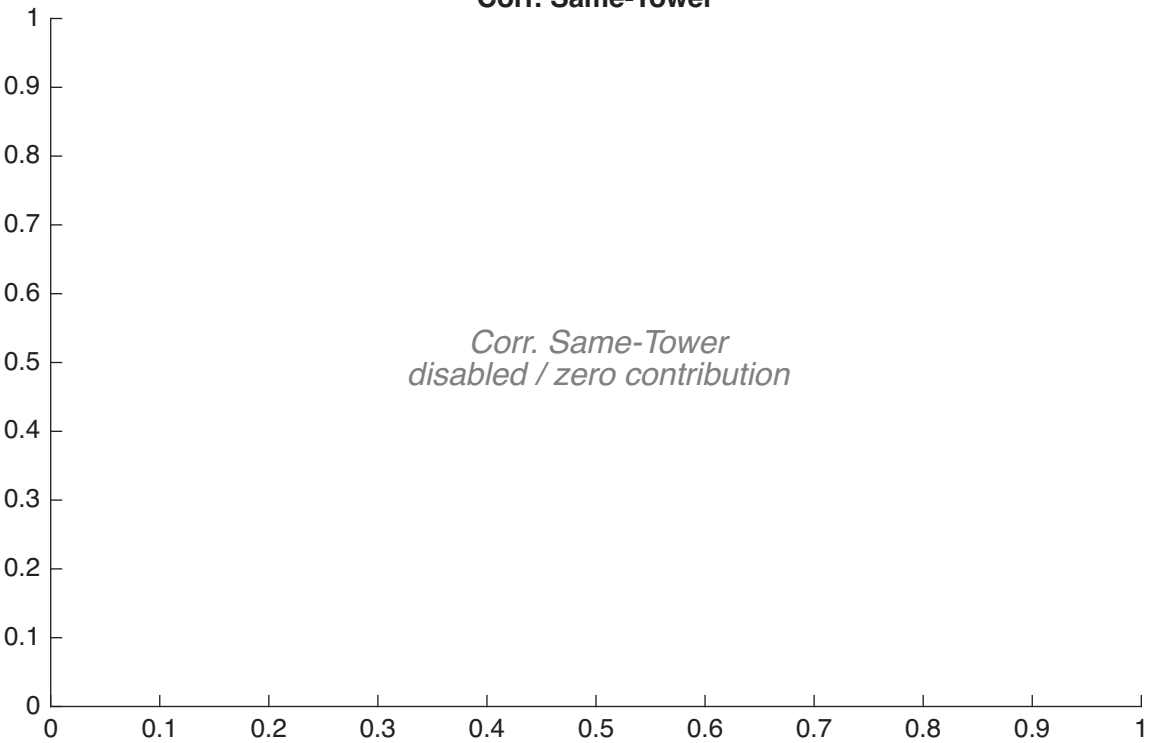
Tower Clock

contribution is numerically zero

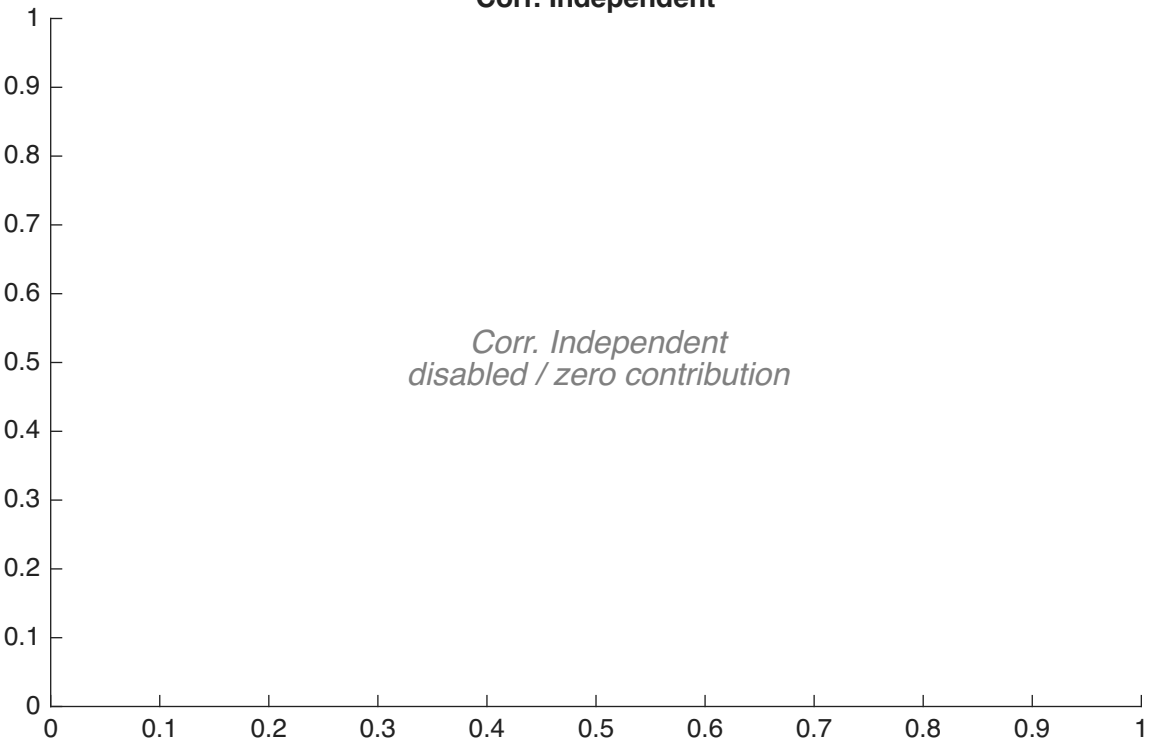
Corr. Common-Mode



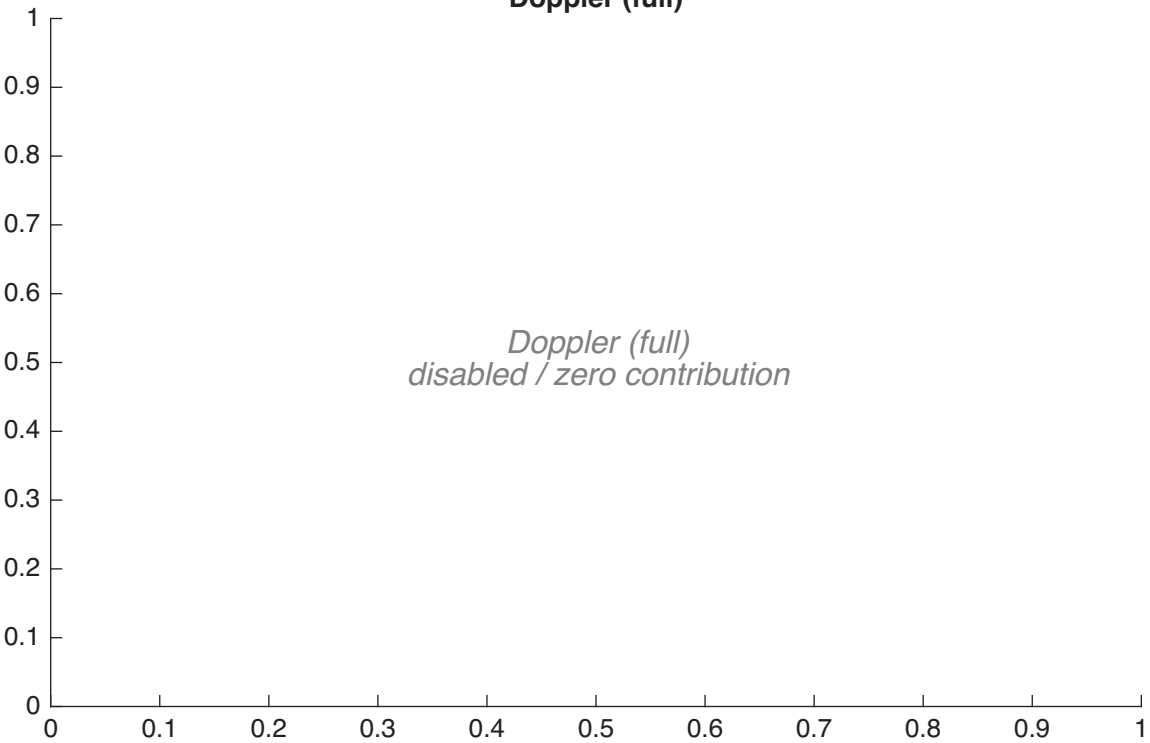
Corr. Same-Tower



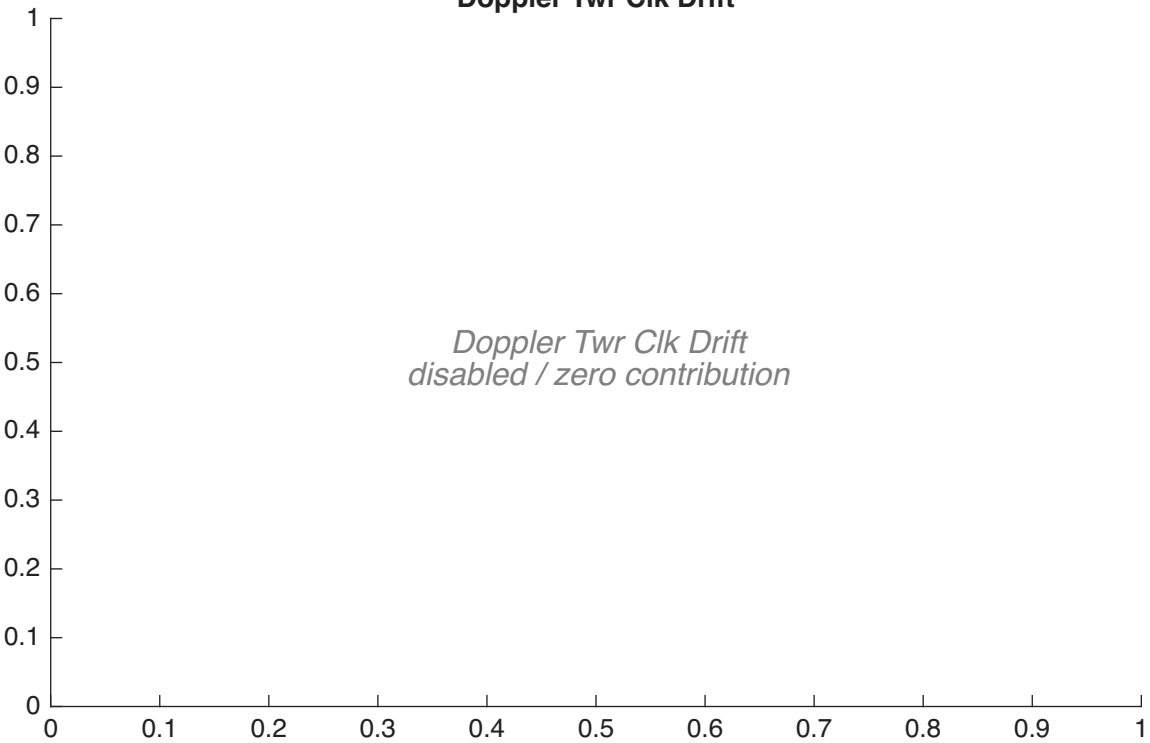
Corr. Independent



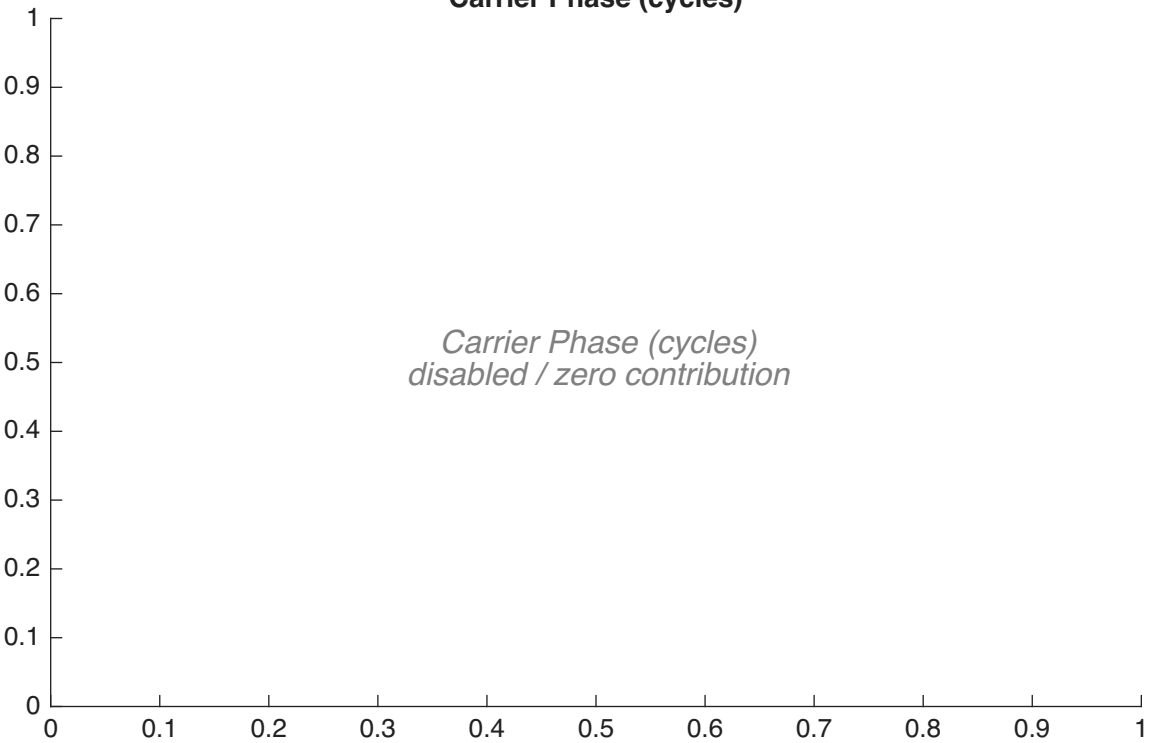
Doppler (full)



Doppler Twr Clk Drift



Carrier Phase (cycles)



Carrier Phase (m)

